



2. CHICKEN POX AND SMALL POX

CHICKENPOX

00:00:18

EPIDEMIOLOGY	CAUSATIVE AGENT	• Varicella zoster virus [human herpes virus – 3]
	INCUBATION PERIOD	• 14 – 16 days
	SOURCE OF INFECTION	• Case of chickenpox (person-to-person/ direct contact, oropharyngeal secretions)
	MODE OF TRANSMISSION	• Droplet & droplet nuclei ◦ Droplet infection: Size of particle: >5micron falls → due to gravity & travels for a distance <1m ◦ Droplet nuclei: Size of particle: <5micron → remains suspended in air & travels for a distance >1m • Vertical transmission leading to congenital varicella
	PERIOD OF COMMUNICABILITY	• 1-2 days before to 4-5 days after appearance of rash
	SECONDARY ATTACK RATE OF CHICKEN POX	• >80% → 85-90%
	M/c LATE COMPLICATION OF CHICKENPOX	• Shingles → reactivation of the virus in 10% cases due to fall in CMI.
	MOST RAPID AND SENSITIVE MEANS OF DIAGNOSIS	• Examination of vesicle fluid under microscope (shows round particles)
	CONGENITAL VARICELLA	• More life threatening if acquired in 1st trimester of pregnancy
	HOST FACTOR	• Age → up to 10 years
	ENVIRONMENTAL FACTORS	• M/c in winters and early spring

CLINICAL FEATURES

PRE-ERUPTIVE STAGE

- Mild to moderate fever
- Malaise
- Chills
- Body ache

ERUPTIVE STAGE

- 1st clinical sign: **appearance of rash**.
- Rash appears on erupts with **1st day of fever**.
- Rash is symmetrical
- It is superficial and area of inflammation is seen around the rash
- Appears on trunk, face, and very few on the extremities
- **Centripetal distribution**: spreads from periphery to centre
- Axilla and mucosal surface are involved



- Palm and soles spared
- Clear fluid vesicle looks like a **dew drop on rose petal appearance**
- **Pleomorphism**: All stages occur simultaneously at the same time
- Unilocular
- With every new crop → eruption fever rises.
- Rapid evolution
 - Scabs (**not infective**) are formed within 4-7 days
- Patient is **non-infectious** once the lesions have crusted

Important Information

Smallpox rash

- Deep seated,
- No area of inflammation around the rash
- **Centrifugal in distribution**
- Palms and soles are affected, and axilla is spared



COMPLICATION WITH CHICKENPOX

- **Varicella pneumonia:** M/c complications in neonates, adults, pregnant woman and immunocompromised patients
- **Reye's Syndrome:** children with clinical chicken pox treated with aspirin
- Immunocompromised individuals can get life threatening complications

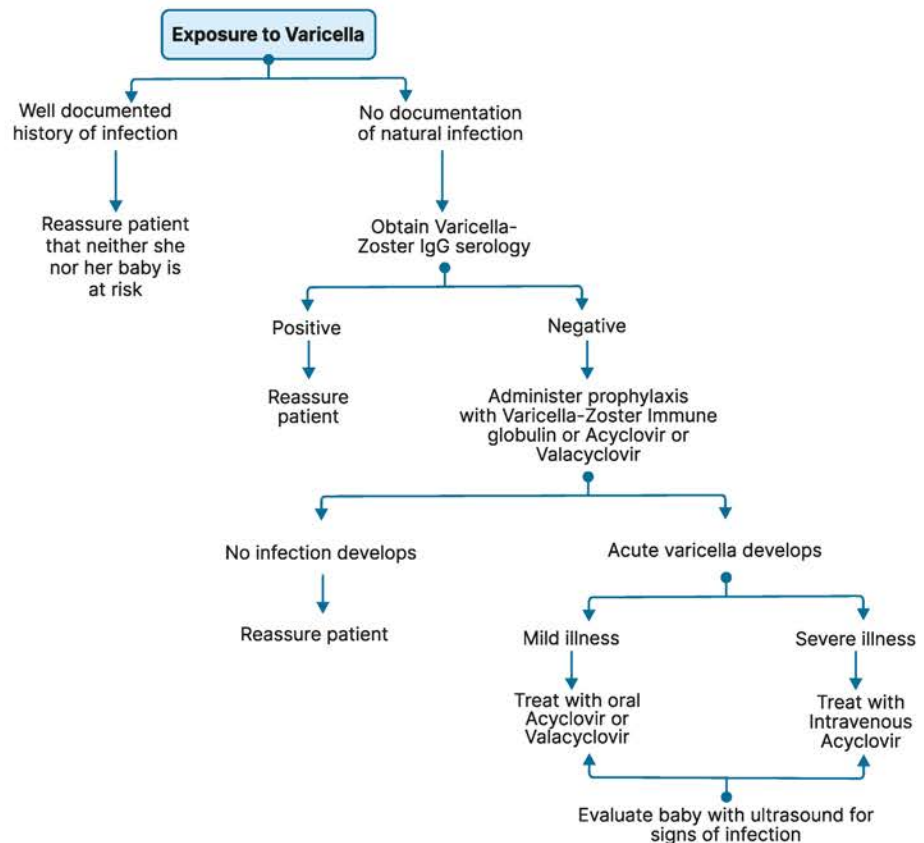
VARICELLA IN PREGNANCY

- Varicella in first 2 trimesters → **Congenital varicella**
 - More life threatening if acquired in 1st trimester
- Varicella at any stage of pregnancy especially in 3rd trimester → **Maternal varicella pneumonia**
- Varicella $\leq$ 5 days antepartum to 2 days postpartum → **Neonatal varicella**

MATERNAL VARICELLA

- Maternal varicella during pregnancy may cause fetal wastage and birth defects such as
 - Cutaneous scars
 - Atrophied limbs
 - **Microcephaly**
 - Low birth weight
 - Cataract
 - Microphthalmia
 - Chorioretinitis
 - Deafness
 - Cerebrocortical atrophy

ALGORITHM FOR THE DIAGNOSIS AND MANAGEMENT OF VARICELLA IN PREGNANCY



PREVENTION

- **Strain:** Oka strain (live attenuated vaccine) for active immunity
 - Not recommended under NIS
- **Type:** Live vaccine
- **Dose:**
 - <13 years (12 months – 12 years): 2 doses: 0.5ml (SC) → gap between 2 doses: 6 weeks to 3 months
 - >13 years: 2 doses 0.5ml (SC) → gap between 2 doses: 4 to 6 weeks

KEY POINT

- Seroconversion: >80%
- Recommended for 12-18 months children who have not had chickenpox

COMBINATION VACCINE

- **MMRV:** Administered to 9 months to 12 years old children
 - 2 doses → 4 weeks apart.
 - Usually, 2nd dose is preferred between 4-6 years of age

KEY POINT

- **Duration of varicella vaccine:** >10 years
- **Adverse reactions:** Tenderness and erythema
- Contraindicated in person with neomycin allergy
- Avoid salicylate for 6 weeks (to avoid Reye's syndrome)

VARICELLA-ZOSTER IMMUNE GLOBULIN (VZIG)

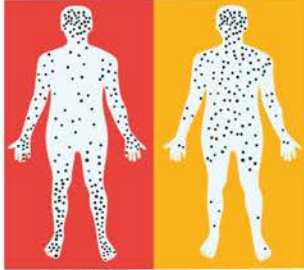
- VZIG is given **within 72 hours** of exposure is recommended for prevention of chickenpox in exposed individuals especially in
 - HIV/AIDS or immunocompromised persons
 - Immunosuppressive therapy patients
 - Congenital cellular immunodeficiency
 - Pregnant females, early neonates, premature infants of low birth weight
 - Dose 1.25 units/kg with maximum of 625 units
- Repeat after 3 weeks if high risk individuals remains exposed

SMALLPOX

00:21:27

FEATURES

- Eradicated on **10th May 1980**
- **Category A bioterrorism agent**
- Human disease
- Caused by variola-virus
- Epidemic disease
- Not associated with lymphadenopathy
- Incubation period is 12 days
- Secondary attack rate = 30-45%
- Transmitted by droplet infection
- Mortality is high

RASH	Centrifugal distribution (From centre to periphery)	Affects extensor surfaces, spares axilla
	Non-pleomorphic	Affects palms and soles
	Deep seated & multilocular	Slow evolution
	No inflammation around vesicles	Scabs form after 10-14 days
CHICKEN POX VS SMALLPOX RASH	 Smallpox Chicken pox	

MCQ

- Q. Which of the following is the next appropriate step for a varicella infected pregnant women with no documentation of natural infection?
- Administer prophylaxis with Varicella Zoster immunoglobulin
 - Prescribe acyclovir/valacyclovir for prophylaxis
 - Evaluate the baby with ultrasound for signs of infections
 - Obtain varicella zoster IgG serology in the women

Ans: (d)

3. RESPIRATORY INFECTIONS, MEASLES, RUBELLA, INFLUENZA AND MENINGITIS



MEASLES

00:00:26

EPIDEMIOLOGY

CAUSATIVE AGENT	SS RNA paramyxovirus (only one serotype exists)
INCUBATION PERIOD	10-14 days
SOURCE OF INFECTION	Case of measles (no subclinical case and no carriers)
MODE OF TRANSMISSION	Droplet ($>5\mu$), Droplet nuclei ($<5\mu$) (airborne)
PERIOD OF COMMUNICABILITY	4 days before and 5 days after the appearance of rash

- Period of isolation : up to 7 days after appearance of rash

Important Information

- Measles **does not show iceberg phenomenon**
 - Because source of infection is only case of measles, no subclinical or carrier states
- Other diseases not showing iceberg phenomenon
 - Tetanus
 - Rabies
- Diseases transmitted by droplet nuclei
 - Measles
 - Chickenpox
 - TB
 - Influenza
 - COVID

SECONDARY ATTACK RATE OF MEASLES	80-85% (one attack gives lifelong immunity)
HOST FACTOR	
AGE	India → 6 months to 3 years
GENDER	M = F
ENVIRONMENTAL FACTORS	• Periodic fluctuation in disease occurrence ◦ Cyclical – epidemics occur in every 2-3 years ◦ Seasonal – winter and early spring
CLINICAL FEATURES	• Prodromal phase (3C) – Cough, Coryza, Conjunctivitis • Eruptive phase – rash

PATHOGNOMONIC CLINICAL FEATURE OF MEASLES	• Koplik spot – bluish white spot-on red base ○ Occurs in buccal mucosa, opposite the lower 2nd molar ○ Koplik spot appears before rash, and disappears as rash emerges
RASH	Maculopapular, confluent rash Retroauricular in origin
M/C COMPLICATION OF MEASLES IN YOUNG CHILDREN	Otitis media
A RARE COMPLICATION OF MEASLES	SSPE – Subacute Sclerosing Panencephalitis • Occurs 7-10 years after infection • Occur in 7/million cases of measles

Important Information

- M/c complication of measles – diarrhea
- M/c cause of death in measles – pneumonia
- M/c fatal complication in measles – pneumonia > encephalitis

DIAGNOSIS

- Clinically
- Gold standard : PCR

TREATMENT

- Symptomatic

CLINICAL PRESENTATION

00:13:00

PRODROMAL PHASE

- 3C's – cough, coryza, conjunctivitis
 - Hyperpigmented line on conjunctiva - **Stimson's line**
- Lacrimation, photophobia, nasal discharge

ERUPTIVE PHASE

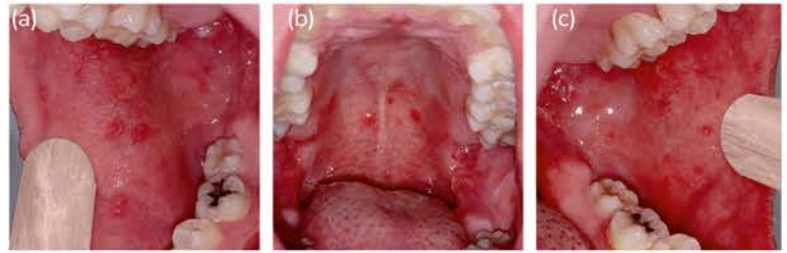
- Red, dusky, maculopapular confluent rash, retro auricular (starts behind ears and spread rapidly)
 - Rash fade in order of appearance
 - A brown discoloration may persist even up to 2 months

Important Information

Rash appears 24-48 hours after kolpik spots

- Severity of the disease depends on extent and confluence of the rash
- In mild measles the rash is not very confluent

- Pathognomonic clinical feature : **Koplik spots**
 - 1 mm grayish white dots with slight reddish areolae
 - Appears in buccal mucosa opposite lower 2nd molar
 - Koplik spot seen in prodromal stage – it disappears when rash emerge



POST MEASLES STAGE

- Peak virus shedding for 2-5 days after rash
- Intense weight loss, fatigue

PREVENTION

00:18:29

ACTIVE IMMUNIZATION

TYPE	Live vaccine
STRAIN	Edmonston Zagreb, Schwarz, Moraten
DOSE	0.5 ml
ROUTE	Subcutaneous (s/c)
SCHEDULE (NIS)	1 st dose – 9 months 2 nd dose – 16 to 24 months

Important Information

- Measles can be administered between 6-9 months in case of
 - Outbreaks
 - Protein energy malnutrition (PEM)
 - Refugees or displaced population
 - Babies of HIV + mother
- These are supplementary doses, they never replace routine dose
- Ideal gap maintained between two doses of measles is **4 weeks**
- Measles is heat and light sensitive vaccine
- Side effects
 - Measles like illness (1-3 days)
 - Incubation period is less than natural incubation period of measles as antibodies formed within 7-10 days after giving vaccine
 - Toxic shock syndrome
- Anyone with **neomycin allergy** should not receive measles vaccine

KEY POINTS

00:24:35

- Should susceptible contacts be given measles vaccine?
 - Can be given within **3 days** of exposure
 - 1st vaccine is given → Ig can be given 2 weeks later

- If Ig given first then vaccine given 8-12 weeks later
→ Never give both together
- Natural incubation (IP) period of measles – 10-14 days
- Post-vaccination antibodies formed – 7-10 days
→ Seroconversion rate following vaccination is faster than natural IP of disease
- **Post- disaster, measles vaccine can be given within 3 days of exposure**
- Does the measles vaccine follow open vial policy?
 - No, because it is a reconstituted vaccine/freeze-dried/lyophilized
 - Add diluents – distilled water/sterile water and used within 4 hours

IMMUNOGLOBULINS

00:30:16

- Passive immunity
- For measles active and passive immunity are not given together
- Measles may be prevented by administering immunoglobulin early in the incubation period
 - Dose : 0.25 mL/kg of body weight, I/m route
 - Schedule : within 3 days of exposure
→ **If Ig is given first → vaccine given 8-12 weeks later**
 - Vitamin A deficiency is encountered along with measles
 - Vitamin A is given along with measles
→ At 9 months 1 lakh IU
→ Every 6 months till 5 years of age 2 lakh IU given

MEASLES ELIMINATION

00:32:43

MEASLES-RUBELLA ELIMINATION CAMPAIGN

- WHO Measles Elimination Strategy: Catch-Up, Keep-Up, Follow-Up

CATCH UP	Nationwide, vaccination campaign targeting all children 9 months to 14 years of age, irrespective of history of measles disease or vaccination status, with one dose of MR • Immunization coverage of ≥95% to achieve elimination
KEEP UP	Routine services aimed at vaccinating > 95% of each successive birth cohort • 1 dose – immunity achieved 85%, • 2 dose – immunity achieved 98%
FOLLOW UP	Subsequent nationwide vaccination campaigns conducted every 2-4 years targeting usually all children born after the catch-up campaign

AMMRS (WHO-UNICEF)

- Accelerated measles mortality reduction strategy
 - High population coverage with 2 doses of measles vaccine or measles-containing vaccine
 - National coverage of >90%
 - District coverage of >80%

RUBELLA : GERMAN MEASLES

00:37:15

- 3-day measles
 - Rash starts on face → spread to trunk → extremities → disappear by 3 days
 - 25% infection can occur without rash
- Acute childhood infection

EPIDEMIOLOGY

AGENT

CAUSATIVE AGENT	SS RNA virus of <i>Togaviridae</i>
INCUBATION PERIOD	2-3weeks (14-21 days), average -18 days
SOURCE OF INFECTION	Case of rubella- one variant exists (Subclinical case), No carriers,
MODE OF TRANSMISSION	Droplet, vertical
PERIOD OF COMMUNICABILITY	7 days before to 7 days after appearance of rash
SECONDARY ATTACK RATE	80% (one attack gives lifelong immunity)

HOST FACTORS

AGE GROUP	3-10 years
GENDER	M = F susceptible
DISTRIBUTION	Developing countries
MODE OF TRANSMISSION	Droplet, vertical
FLUCTUATION IN DISEASE OCCURRENCE	Cyclical distribution : every 5-7 years Seasonal distribution : winters and early spring

CLINICAL FEATURES

00:42:20

PRODROMAL PHASE	• 60-65% infection → asymptomatic • Fever, cough, coryza
LYMPHADENOPATHY	• Posterior auricular and posterior cervical lymph node enlargement as early as 7 days before appearance of rash, • Glands remain enlarged even after 10-14 days of rash
RASH	• Red, maculopapular, discrete, non-confluent, starts from face • Appears 48hours after prodromal stage and fades by 3rd day • 25% infections can be present without rash
COMPLICATION	<i>Congenital rubella syndrome</i>

DIAGNOSIS

- Hemagglutination Inhibition Test (HAI)

TREATMENT

- Symptomatic

PREVENTION

- Rubella vaccine



Measles → Maculopapular, confluent

Rubella → Maculopapular, discrete

CONGENITAL RUBELLA SYNDROME

00:45:48

- Rate of infection and organ damage to fetus is highest in pregnancy
 - Almost 85% of infections occur in the **first trimester**
 - Those babies infected after 20 weeks suffer no major abnormalities
- Classical triad
 - Congenital cataract – most serious complication
 - Sensorineural hearing loss – M/c single complication
 - Cardiovascular defects – PDA > PS > ASD

DIAGNOSIS

00:47:44

- Virus isolation
- **Hemagglutination inhibition test** – M/c
- ELISA tests
- Demonstration of rise in antibody titer between 2 serum samples taken 10 days apart or rubella-specific IgM in single sample accurately confirms recent infection

KEY POINT

- Rubella can be diagnosed at birth
 - At birth, viruses are easily detected in pharyngeal secretions, CSF, multiple organs, urine and rectal swabs
 - Presence of **IgM or IgG** (6 months) indicates presence of congenital infection
 - Viral excretion may last for 12-18 months after birth

BLUEBERRY MUFFIN RASH

00:48:40

- Among the torch group of infections, **cytomegalovirus and rubella** are the m/c viral agents
- Non infective causes of blueberry muffin rash are
 - Congenital spherocytosis
 - Rhesus hemolytic disease
 - ABO blood group incompatibility
 - Anemia caused by twin-to-twin transfusion

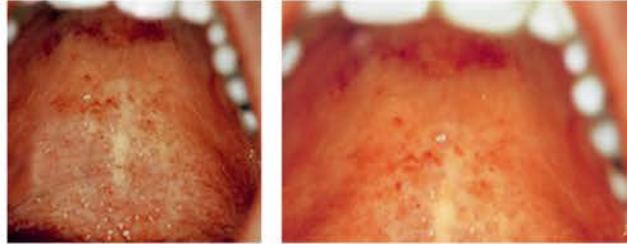


- Blueberry muffin syndrome is a rare neonatal skin disorder characterized by widespread non blanchable, maculopapular lesions of reddish-blue or magenta color
 - Due to persistent dermal erythropoiesis in patients with congenital viral infections

FORCHHEIMER SPOT

00:49:55

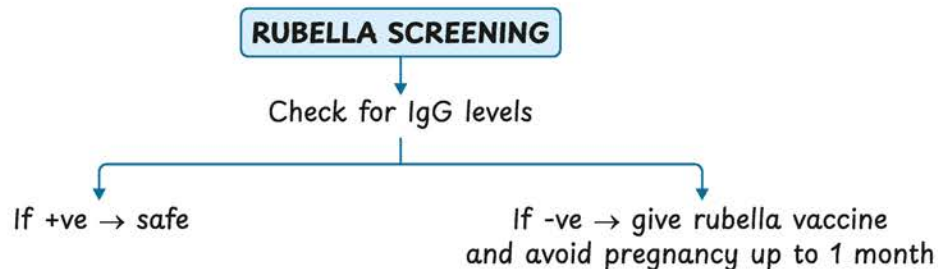
- Discrete rose-colored spot on soft palate that coalesces into red blush and extends over fauces



PREVENTION

00:50:21

- Rubella vaccine
 - Strain - RA27/3
 - Type - Live
 - Dose : 0.5ml,
 - NIS - at 9 months - MR₁; at 16-24 months - Mr₂
 - Route : subcutaneous
 - Recipients advised not to become pregnant over the next 1 month
 - Available as a combined vaccine with MMRV
 - Given between 12 months - 12 years (2 doses)
 - 2nd dose at 5-6 years
- Rubella Vaccination Strategy
 - 1st priority : Women of childbearing age (15-34 or 39/49 years of age)
 - 2nd priority : Interrupt transmission by vaccinating all children 1-14 years age
 - 3rd priority : Routine immunization of all children <1 year (using combined MR or MMR vaccine)



INFLUENZA AND MENINGITIS

00:54:22

MENINGITIS

00:54:38

- Respiratory borne infection
- India → usually reported as public health problem
- Meningitis is endemic diseases in India with 2-10 cases/ lakh population/ year
 - >10 called hyperendemic
 - >100 → Epidemic

EPIDEMIOLOGICAL DETERMINANTS

AGENT	• <i>Neisseria meningitidis</i> ◦ Gram negative coccobacillus ◦ Serotypes: A, C, W₁₃₅, Y, B, X ◦ A, C, W₁₃₅ are the most common
RESERVOIR OF INFECTIONS	• Humans
SOURCE OF INFECTION	• Cases and carriers • If intervention given within 24 hours, diseases is no longer infectious
CARRIERS	• Temporary carriers ◦ Healthy carriers ◦ Convalescent carriers ◦ Chronic carriers ◦ Incubatory carriers
MODE OF TRANSMISSION	• Droplet infection
INCUBATION PERIOD	• 2-10 days
HOST FACTORS	• Age: <2 years- m/c between 3-10 months of age • Equal predilection between males and females
ENVIRONMENTAL FACTORS	• M/C during Winter months • Diseases of lower socio-economic status □ associated with poverty, poor hygiene

CLINICAL FEATURES

00:57:24

- High grade fever
- Stiff neck
- Photophobia
- In untreated cases → Case Fatality rate is up to 5-10%
- In 15-20% cases → patient will report permanent neurological deficit

MENINGOCOCCEMIA

00:58:05

- Fulminant infection: severe form in 20%
- Waterhouse Friderichsen syndrome: Hemorrhagic adrenalitis

DIAGNOSIS

00:58:15

- Microscopy: Bacilli demonstration particularly in CSF
- Agglutination test: serotype specific
- Gold standard: NAT/PCR

PREVENTION AND CONTROL

00:58:52

TREATMENT

CASES	• Penicillin • Ceftriaxone
CARRIERS	• Rifampicin 60mg BD X 2 days
PROPHYLAXIS TO CLOSE CONTACTS	• Ciprofloxacin single dose 500mg or Rifampicin 60mg BD for 2 days

VACCINE

MENINGOCOCCAL CONJUGATE VACCINE

- Type of subunit vaccine

AGE	• Any age group • >2 months to 1 year: 2 doses with a gap of 2 months • 1-30 years: single dose
ADVANTAGES	• Administered to young children • Promotes herd immunity • Longer lasting immunity >5 years
GIVEN TO HIGH-RISK GROUP	• Outbreaks • Pilgrims/travelers- people visiting endemic areas • Post organ transplant • Close contacts of cases

MENINGOCOCCAL POLYSACCHARIDE VACCINE

- Age: >2 years
- Immunity: only for 2 years

TETRAVALENT VACCINE

- Prophylaxis of Meningococcal meningitis
- Effective against the common strains: A, C, Y, W₁₃₅

FLU

01:01:39

- Respiratory borne infection

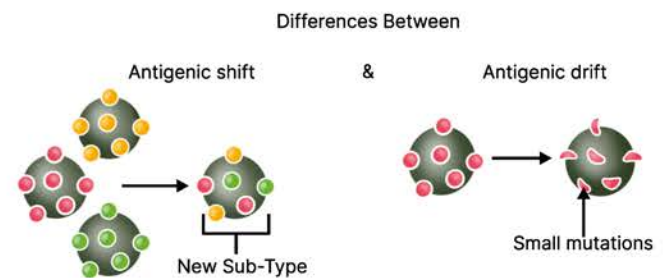
EPIDEMIOLOGICAL DETERMINANTS

AGENT	• Flu virus ○ In India m/c serotype seen is H1N1- H : Hemagglutinin, N- Neuraminidase ○ Other types : H3N2, Influenza type B virus
RESERVOIR OF INFECTION	• Animals, birds and bats
SOURCE OF INFECTION	• Clinical and subclinical case

- Influenza pandemic occurs every 10 years
- Influenza follows iceberg phenomenon

ANTIGENIC DRIFT	ANTIGENIC SHIFT
Minor change within subtype	Major change which creates new subtype
Point mutations	Exchange of gene segments
Occurs in A and B subtypes	Occurs in A subtypes only
Only cause epidemics	May cause pandemic
Example: A/Fujian (H3N2) replaced A/Panama (H3N2) in 2003-2004	Example: H3N2 replaced H2N2 in 1968

- Mode of transmission: droplet and droplet nuclei
- Period of communicability: 1-2 days before to 1-2 days after the onset of symptoms
- Incubation period: **18-72 hours**
- Host factors:
 - Age: any age group
- Environmental factors: winter months, associated with overcrowding, lower socioeconomic status



CHARACTERISTIC FEATURES

- R_0 number- basic reproduction number: 1 case can affect 1-2 cases
 - It's an epidemic if it gives rise to >3 cases
- Case fatality rate (CFR): **1-2%**
- Secondary attack rate: 50-70%

PREVENTION AND CONTROL

TREATMENT

- DOC: Oseltamivir 75mg
 - For treatment: given BD for 5 days
 - For prophylaxis: OD for 5 days

VACCINE

- Live attenuated vaccine- nasal route
- Killed vaccines: A7/California/2009
 - Contraindicated in **egg allergy**



- Q. Which of the following statements regarding measles is not true?
- a. Period of communicability: 14 days from exposure to appearance of rash
 - b. Secondary attack rate for measles: 80%
 - c. One attack gives lifelong immunity
 - d. Agents: single-stranded negative-sense RNA virus

Ans: (a)

- Q. Which of the following stages is characterized by the presence of Koplik spots in measles?
- a. Prodromal stage
 - b. Incubation
 - c. Eruptive
 - d. Convalescent stage

Ans: (a)

- Q. Which strategy is most appropriate for administering measles vaccination to children aged 9 months -14 years to achieve elimination?
- a. Keep up
 - b. Follow up
 - c. Mop up
 - d. Catch up

Ans: (d)

- Q. Which of the following statements regarding German measles (rubella) is not true?
- a. It commonly affects children aged 3-10 years
 - b. The period of communicability extends 7 days before and 7 days after the onset of rash
 - c. German measles shows seasonal distribution, being more common in winter and early spring
 - d. The incubation period is typically 2-3 days

Ans: (d)

- Q. Which age group is prioritized for vaccination against rubella during an outbreak?
- a. Women 15-49 years (non-pregnant)
 - b. Infants
 - c. Women 15-39 years (pregnant)
 - d. Adolescent girls 10-19 years

Ans: (a)

Q. A 5-year-old child presents to a clinic with swelling behind ears. The child developed rashes 10 days ago on the face and spread to the trunk and extremities. On examination, you notice occipital and postauricular lymphadenopathy. The child's mother mentions that he had a mild fever since yesterday. The pediatrician suspects a viral illness. Which of the following is the secondary attack rate of this illness?

- a. 70%
- b. 80%
- c. 40%
- d. 50%

Ans: (b)

Q. Which of the following antiviral drugs is the drug of choice to treat influenza?

- a. Acyclovir
- b. Oseltamivir
- c. Ribavirin
- d. Ganciclovir

Ans: (b)



4. MUMPS AND PERTUSSIS

MUMPS

00:00:29

- Characteristic presentation: Unilateral swelling of parotid gland (parotitis)



EPIDEMIOLGY AGENT

Causative agent	Single stranded RNA virus of Paramyxoviridae family
Incubation Period	14-18 days
Source of Infection	Case of mumps & Subclinical cases (30%-40%)
Mode of transmission	Droplet infection
Period of Communicability	4-6 days before onset of symptoms - 1 week after
Secondary attack rate of Mumps	85%

HOST FACTORS

Age group	5-9 years
Gender	Both genders are equally affected
Immunity	One attack gives lifelong immunity

ENVIRONMENTAL FACTORS

- Periodic fluctuation in the occurrence of disease: Common in winter and early spring

CLINICAL FEATURES, COMPLICATIONS, DIAGNOSIS

00:03:40

CLINICAL FEATURES

- Parotitis- U/L or B/L (rarely) painful swelling of parotid gland
- Can also affect sublingual and submandibular salivary gland
- **Maximum infectivity** - before the onset of parotitis
- No infectivity once swelling subsides

	• Association - Earache on the affected side • Stiffness and difficulty in opening the mouth
	Extra Salivary Gland Manifestation • Orchitis • Oophoritis • Pancreas • Sensorineural hearing loss • Myocarditis • Thyroiditis • Encephalitis
COMPLICATIONS	• M/C complication: aseptic meningitis • M/C complications among young adults: orchitis, oophoritis • Others: pancreatitis, myocarditis, sensorineural hearing loss, thyroiditis • Infection in 1st trimester → spontaneous abortion
MOST RAPID AND SENSITIVE MEANS OF DIAGNOSIS	• Usually done clinically • Isolate mump virus from saliva, urine or CSF. • Positive mumps IgM antibody without mumps immunization in the previous 6 weeks • Seroconversion with 4-fold or greater rise in mumps IgG titer

PREVENTION

00:08:29

- Active immunization
- Mumps vaccine: Live
- Strain: Jeryl Lynn strain
- Other strains:
 - Leningrad-3
 - RIT 4385
 - L-Zagreb
 - Urabe strains
 - Rubini - no longer recommended
- Mumps vaccine can be given as MMR or MMRV
- Route: Subcutaneous
- Dose: 0.5ml, 2 doses
 - 1st dose - 9-12 months
 - Second dose - 5-6 years of age (school going children)

PERTUSSIS

00:10:33

- Life threatening infection particularly for children < 6 months.
- Only way to save the child is DPT vaccine
- Whooping cough
- Hundred Day Cough - paroxysms of cough are followed by an inspiratory whoop (high pitch)

EPIDEMIOLOGY

00:11:01

- **Causative agent**
 - Bordetella Pertussis
 - 5% cases - B. parapertussis
- **Incubation period:** 7-14 days
- **Source of infection**
 - Cases of pertussis
 - Nasopharyngeal secretions, bronchial secretions
- **Mode of transmission:** Droplet nuclei and direct contact
- **Period of communicability:** 1 week after exposure - 3 weeks after the onset of paroxysmal stage
- **Host factors:** Infants and preschool children (< 5 years)
 - Incidence and fatality → higher among Infants < 6 months
- **Gender predisposition:** Females > males
- **Environmental factors:** Winter and early spring
 - Can occur throughout the year (overcrowding)
- **Secondary attack rate of pertussis:** >90% in unimmunized household contacts

COMPLICATIONS, DIAGNOSIS, CONTROL

00:14:14

COMPLICATIONS	• Bronchitis (Secondary bacterial bronchitis) • Bronchopneumonia • Bronchiectasis • Subconjunctival hemorrhages • Epistaxis • Hemoptysis • Cerebral hemorrhage • Apnea and Seizures seen in infants
LABORATORY DIAGNOSIS	• Bacteriological examination of nose and throat secretion → obtained by swab • Culture nasopharyngeal swabs on Bordet-Gengou medium • PCR, immunofluorescence and serological methods
CONTROL	• Isolation and treatment of cases • Vaccination (NIS)
IMMUNITY	• Neither vaccination nor infection confers lifelong immunity

CLINICAL FEATURES

00:16:16

- **Catarrhal stage**
 - Last 10 days
 - Insidious in onset
 - Sneezing, Coryza, Lacrimation
 - Anorexia, Malaise
 - Hacking night cough that starts showing diurnal variation
 - **Most infectious during catarrhal stage**
- **Paroxysmal stage**

- Lasts for 2-4 weeks
- Bursts of rapid, consecutive coughs followed by deep, high-pitched inspiration (whoop), followed by vomiting, cyanosis and apnea
- **Convalescent stage:**
 - Last for 1-2 weeks

TREATMENT

00:17:11

- Isolation of case
- Erythromycin: 30-50 mg/kg QID x 10 days
- For contacts: Erythromycin or Ampicillin x 10 days

PREVENTION

00:17:42

- Pertussis vaccine: DPT or Pentavalent vaccine
- Schedule: 6, 10, 14 weeks
- DPT vaccine
 - Dose: 0.5 ml
 - Route: IM
 - Site: Anterolateral aspect of left thigh
 - 1st booster: 16 - 24 months
 - 2nd booster: 5 - 6 years at arm

	AMOUNT PER DOSE
Diphtheria toxoid	25 Lf
Tetanus toxoid	5 Lf
Pertussis killed acellular bacilli	20,000 million
Aluminous phosphate (adjuvant - ↑ immunogenicity)	2.5 mg
Thiomersal (preservative)	0.01%

- Reactions due to Pertussis component
 - Inconsolable crying- persistent screaming > 3 hrs
 - Hypotonic, hyporesponsive episode
 - Encephalopathy, seizures
- Update: inconsolable crying following DPT/Pentavalent vaccine → no longer an absolute contraindication to subsequent doses

CONTRAINDICATIONS TO PERTUSSIS VACCINE

00:21:21

- Past history of anaphylaxis or vaccine reactions
- Encephalopathy
- Strong family history of epilepsy, convulsions, CNS disorders
- Any concurrent febrile illness

DISEASE	DPT STATUS
Cerebral palsy (non-progressive neurological disorder)	Can give
Active Epilepsy	Cannot give
Epilepsy controlled on anti-epileptic	Can give

Important Information

- DPT vaccine & measles vaccine can result in fever
- Antipyretics - must be given to take home, need based medication



- Q. Which among the following is the most common extra salivary gland manifestations of mumps?
- a. Pancreatitis
 - b. Ovaritis
 - c. Orchitis
 - d. Thyroiditis

Ans: (c)

- Q. All are true about the epidemiological determinants of pertussis except?
- a. The highest mortality is seen in children < 6 months
 - b. The secondary attack rate is > 70%
 - c. The incubation period is 7-14 days
 - d. It can be present throughout the year due to overcrowding

Ans: (b)



5. DIPHTHERIA

EPIDEMIOLOGICAL DETERMINANTS

00:00:23

AGENT

- Gram +ve, non-motile *Corynebacterium diphtheriae*
 - Produces powerful exotoxin
 - Diphtheria is endemic disease in India
- 4 biotypes of diphtheria bacilli
 - Diphtheria gravis : most severe
 - Diphtheria belfanti
 - Diphtheria mitis
 - Diphtheria intermedius

INCUBATION PERIOD

- 2-6 days

SOURCE OF INFECTION

- Case/carrier of diphtheria
 - Carriers → 95% of total infection
 - Incidence of carriers in community : 0.5-1%
 - Nasal carriers are more dangerous than throat carriers
 - Carriers are either healthy, incubatory or convalescent

Important Information

- Immunization does not protect from carrier state

MECHANISM OF TRANSMISSION

00:04:17

- Droplet infection
- Nasopharyngeal secretion
- Cutaneous lesions
- Direct contact
- Contaminated fomites

PERIOD OF INFECTIVITY

00:04:59

- 14-28 days of disease onset in case of untreated cases
- On treatment → Person becomes non-infective within 48 hrs

Important Information

- A case or carrier will be considered non infective when at least 2 cultures obtained from nose and throat 24 hours apart are negative for diphtheria bacilli
- Secondary attack rate is of no relevance

CASE FATALITY RATE

- 10% in untreated children
- Case fatality rate in India 2-3% (with t/t and vaccination in place)

HOST FACTORS

- Age group : 1-5 years
- Both sexes are equally affected

ENVIRONMENTAL FACTORS

- Present throughout the year
- Reported more in winter

CLINICAL TYPES

00:08:47

- Pharyngotonsillar Diphtheria (M/c)
 - Presents with sore throat, difficulty in swallowing, lymph node enlargement, fever, malaise
- Laryngotracheal Diphtheria
- Nasal Diphtheria
- Cutaneous Diphtheria

PSEUDOMEMBRANE FORMATION

- Grayish white to greenish yellow false membrane formed over tonsils, larynx, pharynx
- Well defined edges
- Cannot be wiped, bleeding occurs on wiping it off
- Pseudo membrane formed as a result of bacterial growth, toxin production, necrosis of underlying tissue, host immune response



BULL'S NECK

- Swelling of anterior portion of neck : Submandibular enlargement



COMPLICATIONS

- Toxin is absorbed and results in distant toxic damage
 - CVS : Myocarditis, arrhythmia, CHF
 - Respiratory failure
 - Renal tubular necrosis, proteinuria
 - Neurological
 - Palatal palsy (2nd week) - nasal regurgitation of fluid, nasal intonation of voice
 - Ocular palsy (3rd week) - Deviation of eye
 - Loss of accommodation : Blurring of vision
 - Generalized polyneuritis : 3rd to 6th week
 - Extensive extensive necrosis of heart muscles, liver, kidneys, adrenals
 - Fatty infiltration
 - Parenchymatous degeneration

TREATMENT

00:13:36

- Early detection (Culture methods : swabs from nose and throat & examined by culture methods for detection of diphtheria bacilli)
- Isolation : Cases, suspected cases and carriers should be isolated in a hospital
- Cases : Diphtheria antitoxin given IM or IV immediately after a preliminary test dose of 0.2ml/SC + erythromycin 12-14 days
 - Mild disease : 20000-40000 IU
 - Moderate disease : 40000-60000 IU
 - Severe disease : 80000-100000 IU
- Anterior nasal diphtheria : 10000-20000 IU of diphtheria antitoxin
- Tonsillar diphtheria : 15000-25000 IU
- Pharyngeal/laryngeal diphtheria (<48 hours duration) : 20000-40000 IU
- Nasopharyngeal disease : 40000-60000 IU
- Extensive disease (> 3days) or with brawny swelling of neck : 80000-120000 IU
- For extensive disease add penicillin or erythromycin for 12-14 days

CONTROL OF DIPHTHERIA

00:15:56

CASES	Diphtheria antitoxin After skin sensitivity testing	20,000-1,20,000 IU
	Antibiotics - Penicillin G Oral erythromycin	For 12-14 days
	Serial cultures	Are obtained at regular intervals
	Diphtheria toxoid	Given in convalescent phase only in unimmunized cases
CARRIERS	Oral erythromycin	For 10 -12 days
	Declare as non-infective	When 2 cultures obtained 24 hrs apart from nose/throat are negative for bacilli

CONTACTS (close household contacts)	Antibiotics Oral erythromycin Penicillin G	For 7-10 days	Stop antibiotics after culture reports are negative
	Diphtheria Toxoid	Unknown status or < 3 doses taken	Give diphtheria toxoid and complete series
		> 3 doses and last dose taken > 5 years ago	Give single dose diphtheria toxoid
		> 3 doses and last dose taken < 5 years ago	Continue the series as per NIS

- Antitoxin : 1000 - 2000 units of diphtheria antitoxin

MCQ's



Q. A 17-year-old partially vaccinated girl presents to the emergency department with a low-grade fever, sore throat and difficulty swallowing for the past 3 days. On examination, the girl has a greyish - white membrane covering the tonsils and pharynx and marked neck swelling. A throat culture confirms the presence of *Corynebacterium diphtheriae*. The girl is diagnosed with diphtheria of moderate severity. What is the appropriate dose of Antitoxin for this patient?

- a. 20,000 to 40,000 IU
- b. 40,000 to 80,000 IU
- c. 80,000 to 1,00,000 IU
- d. 10,000 to 20,000 IU

Ans: (b)



6. TUBERCULOSIS

TB INTRODUCTION

- Infectious disease primarily affects lungs → **Pulmonary tuberculosis**
- **Extrapulmonary TB:**
 - Intestine
 - Meninges
 - Bones and joints
 - Skin
 - Lymph glands
- **Bovine TB:** TB of cattle → can be transmitted to man

HIGH YIELD POINTS

- TB → Barometer of social welfare
- 95% of TB cases globally are contributed by middle- and lower-income countries
- M/c cause of death among people living with HIV(PLHIV): TB
- TB associated with smoking: 10 % cases are smokers
- TB associated with diabetes
- Screen for TB as well as NCDs by ASHA: by filling community-based assessment checklist (CBAC)- Rupees 10 for one form filled

EPIDEMIOLOGY

00:04:34

AGENT FACTORS

- **Mycobacterium tuberculosis:** Obligate aerobe and facultative intracellular
- Acid-fast stain-positive
- Other acid-fast stain positive microbes:
 - Nocardia
 - Isospora
 - M. Leprae
- Atypical mycobacteria
 - Photochromogens: M. Kansasii
 - Scotochromogen: M. Scrofulaceum
 - Non photochromogens: M. Intracellulare
 - Rapid growers: M. Fortuitum

DIFFERENCE BETWEEN TYPICAL & ATYPICAL MYCOBACTERIUM

TYPICAL MYCOBACTERIUM	ATYPICAL MYCOBACTERIUM
Highly virulent	Less virulent

Can infect healthy lungs	Capable of infecting diseased lungs (e.g. lungs with congenital deformity, cystic fibrosis)
Can spread directly from human-to-human	No evidence of direct human-to-human spread; they spread from the environment to humans
Are usually sensitive to standard anti-mycobacterial drugs like INH or streptomycin	Are usually resistant to standard anti-mycobacterial drugs
All typical mycobacterial are slow growers	Some atypical mycobacterial are slow grower, and some are rapid growers
All are pigment non-producers	Some are photochromogen; some are scotochromogen, and some are non-chromogen

SOURCE OF INFECTION

- Sputum of a positive case who has either received no treatment or has not been treated completely
- Reservoir: Humans
- Mode of transmission:
 - Droplet- > 5micron
 - Droplet nuclei – airborne
- Bovine TB source: drinking infected milk

HOST FACTORS

- Age: peak age → 15-54 years
- Gender: M > F
- Predisposing factor: Malnutrition

ENVIRONMENTAL FACTORS

- Predisposing factors:
 - Lower socioeconomic status
 - Poverty
 - Malnutrition
 - Overcrowding

PATHOPHYSIOLOGY

00:11:55

PRIMARY TB

- Lesion at hilum of lungs during first infection → Ghon focus
- Ghon focus + lymph nodes: Ghon complex
- Calcification of Ghon complex: Ranke complex
- Outcomes:

Healed TB (95%)	No treatment required
Latent TB (4-5%)	- Person is infected with bacilli, but bacilli is not active - Person is not infectious 5 -10% latent TB cases reactivate to active TB Under stress ↓ Reactivation of Ghon's complex ↓ Spread to the apex (↓sed perfusion) ↓ Results in formation of a cavity ↓ Reactivation TB
Progressive primary TB (1-2 %)	- Immuno-compromised individuals - Local inflammatory response leading to fever, cough, chest pain

SECONDARY TB

- Due to **delayed hypersensitivity response**
- Cavitation at apex with caseous necrosis

COMPLICATION OF TB

00:17:54

- Miliary TB

Important Information

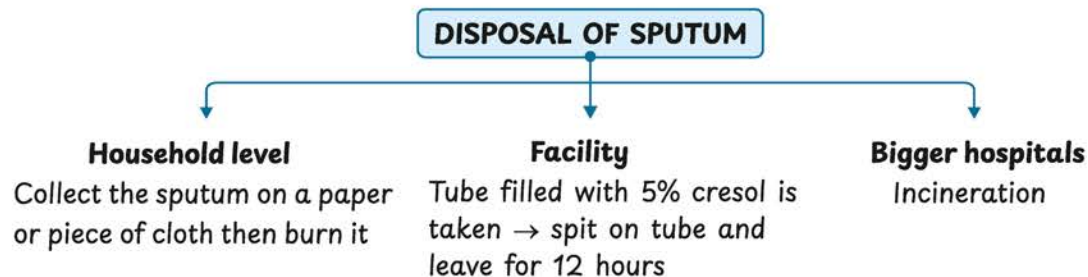
- Any TB-positive person not on treatment can spread disease to 10-15 person/year

DIAGNOSIS

00:19:14

SPUTUM MICROSCOPY

- IOC for mass screening of TB → Detect TB in community**
 - Shows presence of TB bacilli
 - Red acid-fast bacilli on blue background seen
 - Turnaround time: 24 hours
- Low sensitivity
- 2 sputum samples are collected:
 - 1st one is collected on spot
 - 2nd one is collected next morning
- Quantity: > 5 ml
- Even if one sputum sample positive → Person is positive for pulmonary TB
- Disposal of sputum



- Sputum smears are stained for AFB with Ziehl-Neelsen stain
- Decolorizer: Sulphuric acid
- Counterstain: Loeffler's methylene blue

Important Information

- IOC for TB in PLHIV: CBNAAT
- TB suspect at baseline: Sputum microscopy + CBNAAT + chest X ray
- First choice for case finding tool for TB, mass screening of TB: Sputum microscopy

SPUTUM MICROSCOPY REPORTING

- Sputum smear microscopy for tubercle bacilli is positive when there are **at least 10,000 organisms present per ml of sputum**
- Sputum smear examination (ZN staining) by direct microscopy:

Number of AFB	Fields	Report
No AFB	In 100 immersion fields	Negative
1 - 9	In 100 immersion fields	Scanty
10 - 99	In 100 immersion fields	1+
1 - 10	Per-field examination	2+
> 10	Per-field examination	3+

Source: RNTCP lab manual 2009. RNTCP: Revised national tuberculosis control program, AFB: Acid-fast bacilli

MICROSCOPY

- Types
 - Conventional microscopy: use ZN stain
 - LED microscopy: use ZN stain
 - Fluorescence microscopy: use Auramine stain or Rhodamine stain

RADIOGRAPHY

00:27:56

- Useful for diagnosis of sputum-negative pulmonary TB and TB in children

SPUTUM CULTURE

00:28:14

- Solid culture:
 - LJ Medium – **Lowenstein Jensen media**
 - Turnaround time: 84 days
- Liquid culture:
 - **Kirchner's or Middle Brook-based media** – Gold standard for TB
 - Agar-based liquid media is no longer used
 - Turnaround time: 6 weeks

MOLECULAR SUSCEPTIBILITY TESTS FOR DIAGNOSIS OF TB

00:30:19

- Detection and identification of mycobacteria directly from sputum samples
 - CBNAAT (Gene Xpert)
 - Line probe assay (LPA)

CBNAAT

- **Cartridge-based nucleic acid amplification test**
 - IOC for diagnosis and confirmation of TB
- Detects resistance to **rifampicin alone**
- Turnaround time: 2 hours
- **Universal CBNAAT: CBNAAT is done for every TB suspect at baseline**
- IOC for:
 - TB among People living with HIV (PLHIV)
 - Extrapulmonary TB cases
 - Recurrent TB cases
 - Pediatric TB cases
- Indian version of CBNAAT: **TRUENAT**
 - Turnaround time – 1 hour
- False negative result: Low level of detection
- False positive: In recent or recurrent infection (dead bacilli)

LINE PROBE ASSAY

- Detects resistance to both **rifampicin and isoniazid**

PHENOTYPIC TEST

00:35:46

- Based on C14 labelled palmitic acid immunoassay: **BACTEC**
 - Detects CO₂ produced by live organisms
- Measure O₂ consumption by live bacilli: **MGIT**
- Turnaround time:
 - Sputum positive sample: 1 week
 - Sputum negative sample: 2-6 weeks

SUMMARY FOR DIAGNOSIS PROTOCOLS

00:36:39

- For drug-sensitive TB & Drug-resistant TB
 - Sputum, CBNAAT, chest X ray
 - Cytopathology, histopathology, radio imaging support extra pulmonary TB diagnosis

TESTS FOR DETECTING TB INFECTION

00:37:24

1. IGRAS: INTERFERON GAMMA RELEASE ASSAY

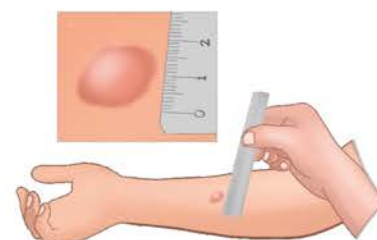
- FDA/ CDC approved IGRA are:
 - QuantiFERON- TB gold In-tube test (QFT-GIT)
 - T-spot
- Used for detecting latent TB also
- No longer done in India

2. MANTOUX TEST/ TUBERCULIN TEST

- Mantoux test (Pirquet test): detect TB infection
- Dose: 1 TU of PPD (purified protein derivative) in 0.1ml injected intradermally on forearm
- Test of prognostic significance
- Limited validity due to lack of specificity
- Result read after 72 hours
- Measure: Induration
- M/c PPD used in India: PPD23 tween 80

Induration >9 mm	Positive (past or current infection with TB)
Induration 6-9mm	Doubtful (M. TB or atypical mycobacterium)
Induration <6mm	Negative (never infected)

- False Positive Tuberculin Test:
 - High coverage of BCG vaccination (protects from extrapulmonary TB)
 - Cross sensitivity to atypical mycobacterium



Measure Induration rather than Erythema

>=5 mm induration:

- HIV positive
- Recent contacts with active TB
- Fibrotic changes on Chest X-ray with prior TB
- Organ transplants and other immunosuppressive patients

≥ 10 mm	- Recent immigrants from high TB-prevalence countries - HIV-negative drug abusers - Mycobacteriology lab personnel - Residents and employees of high-risk occupation groups: nursing homes, hospice centers, health care facilities - Medical conditions which increase TB risk: DM, Silicosis, cancers, blood disorders - Children < 4 years
≥ 15 mm	For people with no risk factors

TWO-STEP TUBERCULIN TESTING

- Done for health personnel or specific groups who are routinely tested
- Some people who were previously infected, as the immune system wanes off, tuberculin test may be negative
- This initial test done may now boost the immune system to react to tuberculin test if given in future
- A subsequent infection may be misinterpreted as a new infection, but due to immune modulation by previous test
- 2 step tuberculin test is recommended for people who are to be tested often and periodically
- First test read within 48-72 hours
 - If positive: person is tuberculin test positive
 - If negative: do second test after 3 weeks
- Second test done after 3 weeks and result read after 48-72 hours
 - If 2nd positive: person infected
 - If 2nd negative: person uninfected

MCQ

Q. A 25-year-old female comes with cough since 5 weeks and blood in sputum. The doctor suspects TB, what is the first investigation to be done as per NTEP guidelines to diagnose TB?

- Rifampicin resistance detection
- Microbiologically confirm TB case
- Sputum AFB + CBNAAT
- Isoniazid resistance detection

Ans: (c)



7. ACUTE RESPIRATORY INFECTION

ACUTE RESPIRATORY INFECTIONS

UPPER RESPIRATORY TRACT INFECTION (URTI)	LOWER RESPIRATORY TRACT INFECTION (LRTI)
• Common cold • Pharyngitis • Epiglottitis	• Laryngitis • Laryngotracheitis • Bronchitis • Bronchiolitis • Pneumonia

ETIOLOGY

00:01:40

COMMUNITY-ACQUIRED PATHOGEN

- Developing countries
 - *S. pneumoniae*, *H. influenza*, *Staphylococcus aureus*
 - Viruses 40%
 - Others : *Mycoplasma*, *Chlamydia*
- Developed countries
 - Bacteria : 5-10%
 - Viruses : RSV, Adenovirus, Parainfluenza, Influenza
 - Others : *Mycoplasma*, *Chlamydia*
- *Streptococcus pneumoniae* is responsible for maximum infection & mortality due to pneumonia among children as well as adults followed by *H. influenzae*

IMNCI : INTEGRATED MANAGEMENT OF NEONATAL CHILDHOOD ILLNESS

00:03:55

- 0 to 5-years are managed under IMNCI
- Disease covered
 - ARI (Pneumonia)
 - Diarrhea
 - Measles
 - Malaria
 - Malnutrition

THE INTEGRATED CASE MANAGEMENT PROCESS

THE INTEGRATED CASE MANAGEMENT PROCESS

Check for danger signs

1. Convulsions
2. Lethargy/unconsciousness
3. Inability to drink/breastfeed
4. Vomiting

Assess main symptoms

1. Cough/difficulty in breathing
2. Diarrhoea
3. Fever
4. Ear problems

Assess

- Nutrition
- Immunization status and
- Potential feeding problems

CHECK FOR OTHER PROBLEMS

Classify the condition of the child and assign to one of the three color codes
and
Identify the treatment actions as per the actions listed in that color band

Urgent referral

1. Pre-referral treatments
2. Advise parents
3. REFER the child
At the referral facility
1. ETAT
2. Diagnosis, treatment and
3. Monitoring and follow up

Treat at the OPD

1. Treat local infection
2. Give oral drugs
3. Advise and teach mother
4. Follow-up

Home Management

- Counsel caretaker on how to
1. Give oral drugs
 2. Treat local infections at home
 3. Continue feeding
 4. Danger signs
 5. Follow-up

PNEUMONIA CLASSIFICATION AND MANAGEMENT

00:08:11

Pneumonia classification for children aged less than 2 months & 2 months to 5 years

FAST BREATHING	
Respiratory rate (RR)	Age
> 60 breaths / min	Up to 2 months
> 50 breaths / min	2-12 months
> 40 breaths / min	1-5 years

- Chest indrawing : when the child breathes in the lower chest wall goes in

CLASSIFICATION

	UP TO 2 MONTHS	2 MONTHS - 5 YEARS
No chest indrawing and no fast breathing	No pneumonia	No pneumonia
Chest indrawing and/or fast breathing	Severe pneumonia	Pneumonia
Danger signs : stridor in calm child, cyanosed, convulsions, high fever, lethargy, not breast feeding	Very Severe pneumonia	Severe pneumonia

T/T

- For no pneumonia : home management (green) in up to 2 months as well as 2 months to 5 years
- Severe pneumonia in up to 2 months : refer to CHC (pink)
 - Give 1st dose of antibiotic – syrup Amoxicillin, keep baby warm & refer
- Pneumonia in 2 months to 5 years : OPD t/t (yellow)
 - Give 1st dose of antibiotic : syrup Amoxicillin, keep baby warm & sent home, follow up in 2 days
- Pneumonia with danger signs in both cases : Urgent referral (pink)
 - Give baby syrup Amoxicillin, keep the baby warm, & refer to CHC

CASE

A 2 year old comes to OPD with mild fever, RR – 38/min, child has chest indrawing, weight is 11 kg, what will be the classification and management?

Ans. Case of pneumonia, as it falls under 2months to 5-year age group, no fast breathing, only chest indrawing present

t/t: Give 1st dose of antibiotic : syrup Amoxicillin, keep baby warm and send home, follow up in 2 days (yellow)

RELEVANCE OF WEIGHT

- Average birth weight of Indian baby is 2.8 kg
- Weight doubles by 5months, triples by 1 year, quadruples by 2 years
- If weight is less it is a case of Severe acute malnutrition (SAM) : referral to CHC as pneumonia becomes severe
 - The conditioning factor for malnutrition is infections

RED FLAGS

00:19:51

- Sign of respiratory distress, nasal flaring & chest indrawing
- Younger than 2 months
- ↓ level of consciousness
- Stridor when calm
- Severe malnutrition
- Associated symptomatic HIV/ AIDS

SAANS

00:20:19

- Social Awareness & Actions to Neutralize Pneumonia: launched in 16th November 2019
- Initiative to reduce deaths due to childhood pneumonia
- 3 interventions : protect, prevent, treat
 - Protect : early initiation of breast feeding
 - Exclusive breast feeding till 6 months
 - Prevent : timely vaccination – PCV, Pentavalent
 - Reduce indoor air pollution by using LPG, & improve ventilation
 - Treat : Mother should be educated to recognize danger signs & get access to timely t/t

CLASSIFICATION AND MANAGEMENT

2 MONTHS – 5 YEARS		
No fast breathing	No pneumonia (cough and cold)	Home management : green
Fast breathing	Pneumonia	OPD management : yellow Give 1 st dose of syrup amoxycillin, keep baby warm
Danger signs and/or chest indrawing and/or SpO ₂ <90%	Severe pneumonia or if baby is malnourished : very severe disease	Urgent referral to CHC : pink

UP TO 2 MONTHS

Danger signs:

Not able to breast feed, lethargic, convulsions, fast breathing, chest indrawing, axillary temperature of baby is $> 37.5^{\circ}\text{C}$, axillary temperature $< 35.5^{\circ}\text{C}$

Possible serious bacterial infection

Referral to CHC: pink



Q. A 7-month-old infant presents to the outpatient department with a cough and fever for two days. She is otherwise healthy. The mother continues to breastfeed her and recently started solid feeds. Her vaccinations are current. On examination, the infant appears irritable and is breathing rapidly. Which among the following is considered fast breathing in an infant between 2-12 months of age?

- a. Respiratory rate > 50/min
- b. Respiratory rate > 60/min
- c. Respiratory rate > 40/min
- d. Respiratory rate > 30/min

Ans: (a)

Q. A 3-month-old baby boy is brought to the pediatric clinic by his parents due to persistent cough, runny nose, and mild fever for 4 days. On examination, the baby is lethargic, irritable, has chest indrawing, and has a respiratory rate of 65/min. The physician tells the parents the baby's symptoms are due to a viral respiratory infection. Which of the following, if present, is NOT considered a red flag sign?

- a. Signs of respiratory distress, nasal flaring & chest indrawing
- b. Severe malnutrition
- c. Stridor when calm
- d. Younger than 4 months

Ans: (d)

8. POLIO



POLIOMYELITIS INTRODUCTION

00:00:36

AFP

- Acute Flaccid Paralysis



ENDEMIC & 8 VULNERABLE COUNTRIES

- **2 endemic countries**
 - Afghanistan
 - Pakistan
 - WPV1 is still circulating in these countries
- **8 Vulnerable countries**
 - Cameroon
 - Sudan
 - Somalia
 - Syria
 - Ethiopia
 - Nigeria
 - Guinea
 - Iraq

POLIO STATUS IN INDIA

- India was declared Polio free on **27th March 2014**
- Last case of Polio reported in India

Polio Timeline in India

WPV2: 24th OCT, 1999 in Aligarh, UP

WPV3: 13th OCT, 2010 in Jharkhand

WPV1: 13th JAN, 2011 in Howrah WB

EPIDEMIOLOGY

00:04:36

CAUSATIVE AGENT

CAUSATIVE AGENT

Enterovirus of picornaviridae family
SS positive sense RNA

	P1	- WPV1 → Most difficult to eradicate, - Still circulating in Afghanistan and Pakistan - Associated with natural outbreaks
	P2	- WPV2 → Maximum Antigenic variation - Easiest to eradicate. - Eradicated on 20 Sep 2015 - Attenuated WPV2 is responsible for CVDPV Circulating vaccine-derived poliovirus
	P3	- WPV3 → Difficult to kill - Attenuated WPV3 is responsible for VAPP → Vaccine-associated paralytic poliomyelitis - Eradicated on OCT. 2019
		WPV can survive in external environment up to 48 hours and can be killed by a disinfecting agent (Fluorine)
MODE OF TRANSMISSION		Feco-oral route (feces and oropharyngeal secretion) - Virus resides in pharynx and peyers patch in GIT - Droplet infection: presence of virus in salivary gland due to ↑ viremia in initial phases
PERIOD OF COMMUNICABILITY		7-10 days before and after onset of symptoms
INCUBATION PERIOD		7-14 days
HOST FACTORS		- Age: < 15 years - M/c in infants - Gender: M > F (3:1)
CLINICAL OCCURRENCE		- M/c are subclinical cases - A single case of polio can give rise to → 1000 subclinical cases in children → 75 subclinical cases in adults - Polio shows iceberg phenomenon
ENVIRONMENTAL FACTORS		- Shows seasonal disease distribution - Seen in rainy, post rainy and winters - Common from September to March
RESERVOIR		Man (No chronic carriers)
RISK FACTORS FOR THE PRECIPITATION OF AN ATTACK		- Fatigue - Trauma - Intramuscular injections - Operative procedures (Tonsillectomy), especially in epidemics of polio - Administration of Alum containing DPT vaccine

CLINICAL PRESENTATION

00:13:17

CLINICAL SPECTRUM	INFECTIONS	REMARKS
Inapparent (Subclinical)	95%	No presenting symptoms, recognizable by isolation or rising antibody titers
Abortive polio (Minor illness)	4-8%	Mild or self-limiting illness, recognizable by isolation or rising antibody titers
Non-paralytic polio	1%	Synonymous with aseptic meningitis
Paralytic polio - Spinal involvement is 80%, - Bulbospinal involvement is 20% - Pure bulbar involvement is rare	<1%	Descending asymmetric flaccid paralysis

DIAGNOSTIC TESTS

00:14:38

STOOL EXAMINATION	- Isolation of wild poliovirus from stool is the recommended method for laboratory confirmation of paralytic poliomyelitis - Recommended in every case of AFP - The virus usually can be found in the faeces from onset to up to 8 weeks after paralysis, with the highest probability of detection during the first 2 weeks after paralysis onset
CEREBROSPINAL FLUID (CSF) EXAMINATION	- Not recommended for purposes of surveillance - Not likely to yield viruses, so collection is not recommended for culture - CSF cell count, gram stain, protein, and glucose may be very useful in eliminating other conditions that cause AFP
THROAT EXAMINATION	- Not recommended for purposes of surveillance - Not as likely as a stool to yield virus, and thus specimen collection from this site is not recommended
BLOOD EXAMINATION	- Not recommended for purposes of surveillance - Not likely to yield virus, and current serologic tests cannot differentiate between wild and vaccine virus strains - Interpretation of the serologic data can often be misleading - Collection of blood specimens for culture or serology is not recommended.

VACCINATION

00:16:05

NIS		OPV	IPV
	At birth	OPV ₀ given up to 15 days	
	6 weeks	OPV ₁	fIPV ₁

	10 weeks	OPV ₂	
	14 weeks	OPV ₃	fIPV ₂
	9 months		fIPV ₃
	16-24 months	OPV _{booster}	
	Delayed immunization	Up to 5 years	Up to 1 year
DIFFERENCE BETWEEN OPV & IPV		OPV (Sabin)	IPV (Salk)
	Type of vaccine	Live	Killed
	Mode of administration	Oral – 2 drops	ID fIPV-0.1ml
	Type of immunity	Humoral + Intestinal (local)	Humoral
	Prevention of	Paralysis + intestinal re-infection	Paralysis
	Control of epidemics	Effective	Not effective
	Storage & transport	Stringent condition. Require sub-zero temperatures	Less stringent conditions
	Shelf life	Shorter	Longer
	VAPP	1 per 1 million case	Zero incidence
	CVDPV	Yes	No
	Important Information • Trivalent OPV contains attenuated forms of WPV1, WPV2 & WPV3 ◦ An attenuated form of WPV2 leads to CVDPV ◦ An attenuated form of WPV3 leads to VAPP but is rare • Therefore, SWITCH from trivalent OPV to bivalent OPV was proposed in April 2016 • Bivalent OPV does not contain WPV2 • IPV is not associated with paralytic attacks		
INACTIVATED (SALK) POLIO VACCINE (IPV)	COMPONENTS		STRENGTH
	Poliovirus type 1		20 D antigen units
	Poliovirus type 2		2 D antigen units
	Poliovirus type 3		4 D antigen units

Advantages of IPV

- Safe in immunodeficiency disorders
- Safe in persons on radiation therapy/corticosteroid therapy
- Useful in those over 50 years of age
- Safe during pregnancy
- No risk of Vaccine-associated paralytic polio (VAPP)

IPV is unsuitable in epidemics

- Immunity is not rapidly achieved as > 1 dose required
- Injections can precipitate paralysis during epidemics

ORAL (SABIN) POLIO VACCINE (OPV)

- Live attenuated 'trivalent' vaccine
 - Contains 3 strains of poliovirus
- Bivalent OPV is present in NIS and pulse polio

Mechanism of action:

- **Primary Multiplication:** Intestinal epithelial cells
- **Secondary Multiplication:** Peyer's patches (leads to viremia)
- Induces 'both systemic and local immunity' (Nasal & duodenal IgA, Serum IgM, IgG, IgA)

Composition of OPV

COMPONENTS	STRENGTH
Poliovirus type 1	3 lac TCID 50
Poliovirus type 2	1 lac TCID 50
Poliovirus type 3	3 lac TCID 50

Advantages of OPV

- Easy to administer
- Induces both humoral and systemic immunity
- A single dose produces substantial immunity
- Vaccines spread immunity to others by excretion of virus → herd immunity
- Inexpensive
- Useful in controlling epidemics



Q. All are true about the Sabin vaccine except?

- a. It is a live attenuated vaccine
- b. The mode of administration is via the oral route
- c. The shelf life is longer
- d. It required sub-zero temperatures for storage and transport

Ans: (c)

Q. What is the fractional dose of Inactivated Polio Vaccine (IPV) administered under the National Immunization Schedule?

- a. 0.5 ml
- b. 0.2 ml
- c. 0.1 ml
- d. 1 ml

Ans: (c)

Q. Most natural outbreaks of polio are due to?

- a. Type I virus
- b. Type II virus
- c. Type III virus
- d. All of the above

Ans: (a)



9. HEPATITIS

COMPARISON IN TYPES OF HEPATITIS

00:00:10

	HEPATITIS A	HEPATITIS B	HEPATITIS C	HEPATITIS D	HEPATITIS E
CLASSIFICATION	Picornavirus	Hepadnaviridae	Flavivirus	Delta virus	Calicivirus
GENOME	ss-RNA non-enveloped virus	ds-DNA enveloped virus	ss-RNA enveloped virus	ss-RNA enveloped virus	ss-RNA non-enveloped virus
SOURCE	Feces	Blood/body fluids	Blood/body fluids	Blood/body fluids	Feces
TRANSMISSION	Feco-oral route	Parenteral, perinatal, sexual	Parenteral, blood transfusion, organ donation	Parenteral, sexual	Feco-oral route
IP	15-45 days	45-190 days	30-120 days	30-90 days	25-45 days
ONSET	Abrupt	Insidious	Insidious	Abrupt	-
SEVERITY	Mild	Chronic carriers	Subclinical infection	Co-infection with Hep B	Mild (except pregnant women)
FULMINANT HEPATITIS	Rare	Very rare (5-10%)	Rare	Occasional	Rare
SYMPTOMS	Vomiting, Diarrhoea, Jaundice, Fever, Malaise	Symptoms in 10-15% can have serum sickness (rash, arthralgia)	20% symptomatic	Same as A	Same as A
CHRONICITY (%)	0%	5-10%	80%	5%	0%
CARRIER STATE	No	Yes	Yes	Yes	No

ASSOCIATED WITH BLOOD TRANSFUSION	Very rare	5-10%	Almost negligible	Rare	Rare
DIAGNOSIS (SEROLOGY)	Anti-HA antibody formed 2 weeks after raised liver enzymes	Viral markers	Antibody to Hepatitis C virus persists for 6 months	RT-PCR	RT-PCR, IgM, IgG
POST-EXPOSURE PROPHYLAXIS	IGs + HAV vaccine	IGs + Vaccine	Antiviral agents	None	Unknown
ASSOCIATED WITH CIRRHOSIS	No	Yes	Yes (M/c cause of cirrhosis is Hepatitis C)	Yes	No
ASSOCIATED WITH HCC	No	Yes	Yes	Yes	No

Important Information

- **Hepatitis A vaccine**
 - Live vaccine – SC, Single dose
 - Killed vaccine- 2 doses (6-12 months apart) >1 year
- **Hepatitis A Ig** → 90% effective if given within 14 days of exposure
 - Dose : 0.05-0.1ml/kg

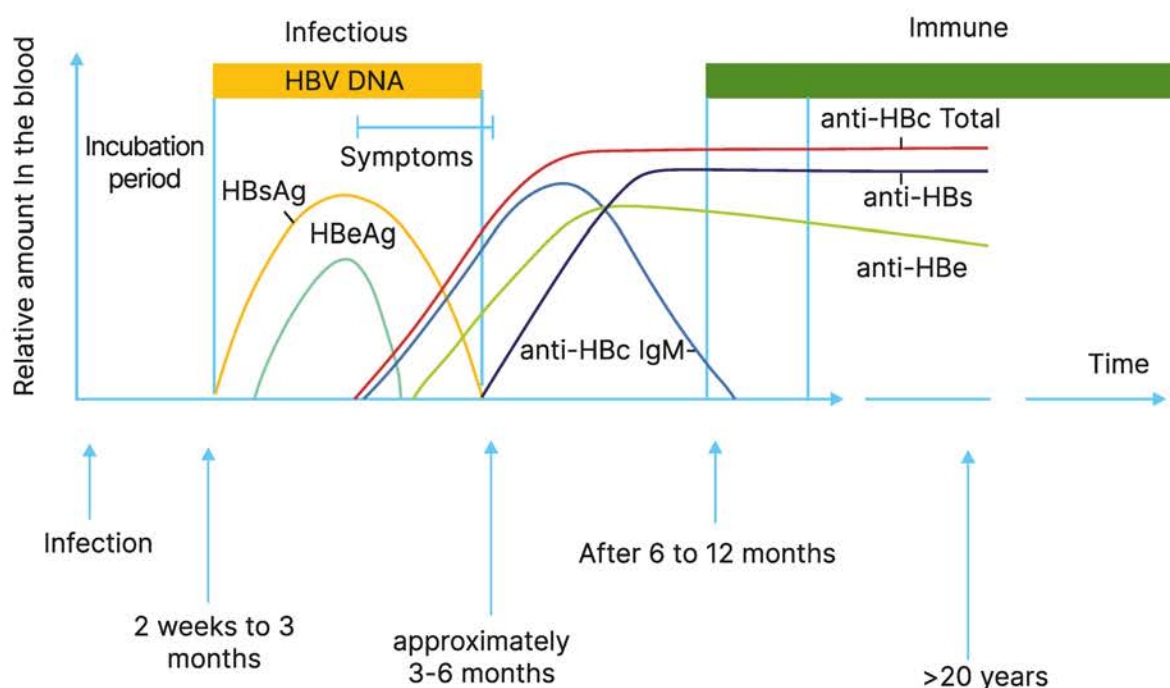
MARKERS OF HEPATITIS B INFECTION

00:09:31

ANTIGEN	HBsAg (Australian Ag)	1 st epidemiological marker to be detected signals acute or chronic infection if it persists for > 6 months or carrier state
	HBc Ag	The HBcAg is a marker of the infectious viral material, and it is the most accurate index of viral replication. Its specific antibody is anti-HBc
	HBeAg	Hepatitis B e antigen appearing during weeks 3 to 6 indicates an acute active infection at its most infectious period & means that the patient is infectious. Persistence of this virological marker beyond 10 weeks shows progression to chronic infection & infectiousness, replication, & the patient is highly infectious
ANTIBODY	anti-HBc	IgM anti-HBc Ag signals new infection IgG anti-HBc Ag old infection
	anti-HBe	Good prognostic feature signals low infectivity

	anti-HBs	Recovery from HBV infection, development of immunity & cure. Its appearance 1 to 4 months after the onset of symptoms indicates clinical recovery & subsequent immunity to HBV
HEPATITIS B X ANTIGEN	• Detected in the blood of HBeAg positive patient • In both acute & chronic hepatitis • HBxAg is a transcriptional activator • Not bind to DNA • Its specific antibody is anti-HBx	
HBV DNA	• Detected by PCR as soon as 1 week after initial infection • Only performed for research purposes or to detect mutants	
HBV DNA POLYMERASE TESTS	• Presence of HBV DNA polymerase • Detectable within 1 week of initial infection • Only performed for research purposes	

HBV antigens and antibodies in the blood



ASSAY RESULTS			INTERPRETATION
HBsAg	Anti-HBs	Anti-HBc	
+	-	-	Early acute HBV infection
+	+/-	+	- Acute or chronic HBV infection, Differentiate with IgM-anti-HBc - Determine the level of infectivity with HBeAg or HBV DNA

-	+	+	Indicates previous HBV infection & immunity to hepatitis B
-	-	+	Possibilities include past HBV infection; low-level HBV carrier; time span between disappearance of HBsAg & appearance of anti-HBs; or false-positive or nonspecific reaction
-	-	-	Another infectious agent, toxic injury to the liver, disorder of immunity, hereditary disease of the liver, or disease of the biliary tract
-	+	-	Vaccine-type response

VACCINE

00:15:26

HEPATITIS B VACCINE	- Recombinant monovalent combination or with other vaccines - Hepatitis B vaccine is prepared using HbsAg - NIS: given at birth, IM anterolateral aspect of the thigh can be given up to 24 hours - And in the form penta at 6 weeks, 10 weeks, & 14weeks - For adults : Dose : 20 mcg (1ml) at 0 dose, 1 month & 6 months as IM injections HIGH-RISK INDIVIDUALS - IV drug users - High-risk sexual behaviour - Blood transfusion - Health care workers - Organ transplant recipients
HEPATITIS B POST-EXPOSURE PROPHYLAXIS	- Acute exposure to HBsAg-positive blood (accidental exposure) - HBIG: 0.05 -0.07 ml/kg as soon as possible or within 6 hrs or upto 24 - 48 hours - Administer Hep B vaccine 1ML IM at 0, 1, 6 months - Vaccine not required in cases of - If an exposed person is HBsAg positive - History of previous infection in an exposed person with HBV & anti-HBs antibody >10mIU/ml
MOTHER TO CHILD TRANSMISSION	- Mother is a chronic carrier - Mother suffers from acute infection during 1st trimester - During passage through the birth canal - Post-natal period due to close contact - To newborns, the Hepatitis B vaccine & Hepatitis B immunoglobulin are given as soon as possible after birth

MCQ's



Q. The 1st antibody to appear against hepatitis B is :

- a. Anti HBs
- b. Anti-HBe
- c. Anti-HBc
- d. Anti-HBm

Ans: (c)



10. DIARRHEA

DIARRHEA

00:00:25

- Defined as **passage of 3 or more loose watery stools in 24-hour period**
- Recent change in consistency and character of stool is more important than the number of stools
- Complications : Dehydration and electrolyte imbalance

CLINICAL TYPES OF DIARRHEA

00:02:08

ACUTE WATERY DIARRHEA

- Lasts several hours to days
- Dehydration, weight loss occurs if it persists
- Organisms
 - *V. cholerae*
 - *E. coli*
 - **Rotavirus** (M/c cause of diarrhea among infants)

ACUTE BLOODY DIARRHEA

- AKA : Dysentery – (blood in stools)
- M/c organism responsible for bloody diarrhea is **Shigella**
- Damage of the intestinal mucosa → Sepsis and later malnutrition

PERSISTENT DIARRHEA

- Diarrhea lasts for > 14 days
- Seen in **HIV +ve**
- Leads to malnutrition and weight loss

DIARRHEA WITH SEVERE MALNUTRITION

- Seen in children suffering from **Severe acute malnutrition** (SAM) : Marasmus and Kwashiorkor
- Leads to dehydration, systemic infection and heart failure

COMPOSITION OF ORS SOLUTIONS

00:05:05

CONTENTS	REDUCED OSMOLARITY (MMOL/L)
Glucose	75 mmol/l
Sodium	75 mmol/l
Chloride	65 mmol/l

Potassium	20 mmol/l
Citrate	10 mmol/l
Osmolarity	245 mmol/l

- ORS should be consumed within 24 hrs

REDUCED OSMOLARITY ORS	g/L
Sodium chloride	2.6
Glucose, anhydrous	13.5
Potassium chloride	1.5
Trisodium citrate, dihydrate	2.9
Total weight	20.5

ReSoMal

00:06:44

- T/t of dehydration in **severely malnourished children**
- Composition : WHO ORS packet+ 2 L water + 50 gm sucrose + 40 ml (or 45 mL of KCl solution, Mg, Zn, Cu)

COMPOSITION	ReSoMal (mmol/L)	STANDARD ORS (mmol)	REDUCED OSMOLARITY ORS
Glucose	125	111	75
Sodium	45	90	75
Potassium	40	20	20
Chloride	70	80	65
Citrate	7	10	10
Magnesium	3	---	---

KEY POINT

COMPOSITION	ReSoMal (mmol/L)	STANDARD ORS (mmol)	REDUCED OSMOLARITY ORS
Zinc	0.3	---	---
Copper	0.045	---	---
Osmolarity (mOsm/L)	300	311	245

- ReSoMal has additional elements : **Mg, Zn, Cu**

SUPER ORS

00:08:40

- Special ORS containing complex sugars
- Food based(rice-based) + Glycine, alanine + Glucose Polymers
- Advantages
 - Rehydration
 - Reduces stool output, frequency of stools, and duration of diarrhea
 - Provides extra 180 kcal/L calories → weight gain
- Limitation
 - Short shelf life (<10 hours), needs to be freshly prepared 2-3 times a day

IMPORTANCE OF EACH COMPONENT OF ORS

00:10:27

- Glucose
 - Facilitates absorption of sodium & hence water
- Potassium
 - Maintains electrolyte balance
 - Na⁺ & K⁺ replaces loss (during diarrhoea & vomiting)
- Trisodium citrate
 - Improves shelf life of ORS (makes ORS stable)
 - Allows intestine to absorb sodium & water
 - Corrects acidosis (occurring in diarrhoea & dehydration)
 - Reduces stool output
- Reduces stool output by 25% (in comparison to original formula)
- Reduces vomiting by 30 %
- Reduces the need for IV therapy by > 30 %

MANAGEMENT OF DIARRHEA

00:12:06

- 3 categories
 - No signs of mild dehydration – Plan A
 - Some signs of dehydration – Plan B
 - Severe dehydration – Plan C

DEGREE OF DEHYDRATION

00:12:37

FACTORS	MILD	MODERATE	SEVERE
General Condition	Well, alert	Restless, thirsty, irritable	Drowsy, cold extremities, lethargic
Eyes	Normal	Sunken	Very sunken, dry
Anterior fontanelle	Normal	Depressed	Very depressed
Tears	Present	Absent	Absent
Mouth + tongue	Moist	Sticky	Dry

Skin turgor	Slightly ↓	↓ (1-2 sec)	↓↓ (> 2 sec)
Mouth + tongue	Moist	Sticky	Dry
Skin turgor	Slightly ↓	↓ (1-2 sec)	↓↓ (> 2 sec)
Thirst	Normal	Drinks eagerly, thirsty	Unable to drink or drinks poorly
Pulse (N=110-120 beat/min)	Slightly ↑	Rapid, weak	Very rapid, impalpable
BP (N=90/60 mm Hg)	Normal	↓	↓↓ (unrecordable)
Respiratory rate	Slightly ↑	Rapid	Deep, rapid
Urine output	Normal	Reduced	Markedly reduced
Treatment	Plan A	Plan B	Plan C

PLAN A

00:16:37

TREAT DIARRHEA AT HOME

- For no/mild dehydration
- Management is done according to IMNCI: Color code for home management is **green**
- Counsel the mother on the 4 rules of home treatment:
 - Give extra fluid
 - Give zinc supplements (age 2 months up to 5 years)
 - Continue feeding
 - When to return
- Show the mother how much fluid to give in addition to the usual fluid Intake
 - **Up to 2 years** : 50 – 100 ml after each loose stool
 - **2 years or more** : 100 – 200 ml after each loose stool
- Give Zinc : 20 mg tablet
 - 2 months to 6 months : ½ tablet → 10 mg for 14 days
 - 6 months and more : Full tablet → 20 mg for 14 days
 - Reduces duration of diarrhea

PLAN B

00:18:44

TREAT SOME DEHYDRATION WITH ORS

- Given in moderate dehydration
- In the clinic, give recommended amount of ORS over 4-hours period
- Determine amount of ORS to give during first 4 hours

WEIGHT	<6kg	6-<10 kg	10-<12kg	12-19kg
AGE	Up to 4 months	4 months up to 12 months	12 months up to 2 years	2 years up to 5 years
In ml	200-450	450-800	800-960	960-1600

- For infants under 6 months who are not breastfed, also give 100-200 ml clean water during this period if you use standard ORS

PLAN C

00:19:54

TREAT SEVERE DEHYDRATION QUICKLY

START HERE

Can you give intravenous (IV) fluid immediately?

YES →

- Start IV fluid immediately.** If the child can drink, give ORS by mouth while the drip is set up. Give 100 ml/kg Ringer's lactate solution (or, if not available, normal saline), divided as follows:

Age	First give 30 ml/kg in:	Then give 70 ml/kg in
Infants (under 12 upto 5 months)	1 hour	5 hours
Children (12 months upto 5 years)	30 minutes	2 ½ hours

- * Repeat once if radial pulse is still very weak or not detectable.
- Re-assess the child every 1-2 hours. If hydration status is not improving. Give the IV drip more rapidly
- Also give ORS (about 5 ml/kg/hour) as soon as the child can drink, usually after 3-4 hours (infants) or 1-2 hours (children)
- Re-assess an infant after 6 hours and a child after 3 hours. Classify dehydration. Then choose the appropriate plan (A, B or C) to continue treatment



- Q. A 2-year-old child is brought to the emergency department with complaints of the frequent passage of watery stools for 2 days. On examination, the child is well alert, eyes and anterior fontanelle are normal and thirst is normal. Vital examination shows a slight increase in pulse rate and respiratory rate. What is the next step in the management of this child?
- Treat diarrhea at home with extra fluids and zinc supplements
 - Treat with ORS
 - Treat with the administration of IV fluids
 - No intervention required

Ans: (a)

- Q. A 5-year-old child presents to the clinic with a 2-day history of watery diarrhea, vomiting, and mild dehydration. Upon examination, the child appears lethargic with dry mucous membranes. The physician diagnoses acute gastroenteritis and recommends oral rehydration therapy. What is sodium concentration of reduced osmolarity ORS?
- 75 mmol/l
 - 65 mmol/l
 - 20 mmol/l
 - 10 mmol/l

Ans: (a)

11. CHOLERA

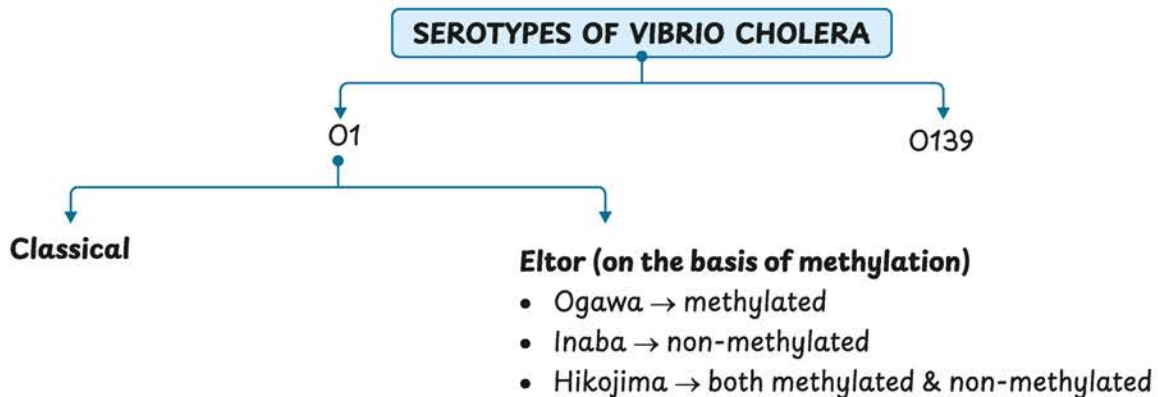


EPIDEMIOLOGICAL DETERMINANTS

00:01:52

AGENT

- **Vibrio cholera**: Gram -ve, curved, rod-shaped, motile bacilli



- M/c in India: **ELtor serotype, Ogawa biotype**

Classical O1	ELtor O1
Shorter epidemic	Longer epidemic
↑ secondary cases	↓ secondary cases
Chronic carrier ↓	Chronic carrier ↑
Less resistant	More resistant

- Vibrio cholera can be destroyed by following physical factors
 - Heat (30-40°C for 30 minutes)
 - Drying
 - Bleaching powder (5-6 mg/L)
- Cholera AKA: **blue death**
 - Associated with cyanosis, gray-blue tinge due to dehydration

SOURCE

00:06:24

- Case or carrier of cholera
 - 75% of people infected with cholera do not develop symptoms
→ **Subclinical cases/ carriers**
 - For every 1 case of cholera → 50-100 subclinical cases
- Infective material: Stool or vomitus of case/carrier
- Carriers: infectivity period

Case of cholera	8-10 days
Incubatory carrier	1-5 days
Healthy carrier	10 days
Convalescent carrier	2-3 weeks
Chronic carrier(rare)	months – years

- Reservoir of Infection: **Humans** (only reservoir)
- Period of communicability: 8-10 days to months/years
- Mode of transmission: Contaminated water, food, direct contact
- Incubation period: 1-2 days
- Quarantine period: 5-7 days for healthy contacts

ENVIRONMENTAL FACTORS

- Rainy season
- Overcrowding
- Lower socioeconomic status

CLINICAL FEATURES

00:09:40

- 75% do not develop disease
- 5-10% mild cases
- 15-20% severe dehydration
- **Stage of evacuation:** Profuse, painless diarrhea (**rice water stools**)
- **Stage of collapse:** Severe diarrhea and dehydration
- **Stage of recovery**

DIAGNOSIS

00:10:30

- Collect stool sample
- In community → Collect water sample
- Stool collection and transportation
 - **Mc Cartney bottles** for transport of stool
 - Transport holding media: **Venkatraman Ramakrishna (VR) media**
 - Rectal swabs transported in alkaline peptone water or **Cary Blair media**
- Culture methods
 - Enrichment: Inoculation onto peptone water **tellurite media**
 - **Hanging drop preparation:** Demonstrates scintillating movements
 - Darkfield illumination: **shooting stars in dark sky**
 - Selective media: **Thiosulfate citrate bile salt sucrose agar (TCBS)**
- Serological tests

TREATMENT

00:12:05

- Isolation: Till 2-3 negative stool culture reports obtained
- Cholera is a notifiable disease
- Degree of dehydration: plan A, plan B, plan C

- ORS + Zinc (promotes epithelialization): 10 - 20 mg for 14 days
- Antibiotics: Doxycycline, tetracyclines, fluoroquinolone
 - For children & pregnant women: Erythromycin
- Sanitation measures:
 - Concurrent and terminal disinfection of stool and vomitus with **cresol**
 - Safe water supply
 - Safe excreta disposal

CONTROL FOR CHOLERA

00:14:57

- Chemoprophylaxis: **Tetracycline** x 3 days
 - Adults 500mg BD x 3 days
 - Children (4-13 years) 125 mg BD X 3 days
 - Children (0-3 years) 50 mg BD x 3 days
 - Other alternative: Doxycycline 300mg single oral dose

IMMUNIZATION

00:15:27

- Parenteral vaccine: Heat killed; phenol preserved vaccine not used
- Oral vaccines

DUKORAL	SANCHOL AND MORCVAX
• For O1 type- monovalent • Gastric buffer used as it contains Cholera toxin B subunit • Dose: 2 doses; 1-6 weeks apart • Children: 3 doses; 1-6 weeks apart • Children < 2 years not given • Effectivity starts after 7 days and lasts for 2 years	• For O1 and O139 → Bivalent • Only oral suspension, no gastric buffer used • 2 doses; 2 weeks apart • Not given to children <1 year • Effectivity starts 7 days after and lasts for 3 years

Important Information

- **EUVICHOL**: Recently approved, more stable vaccine

WHAT SHOULD BE DONE IN CHOLERA OUTBREAK?

00:18:11

- Cholera case confirmation: Assess population coverage
- Outbreak response team: Epidemiologist/microbiologist (should have stock like electrolytes, disinfectant)
- Isolation: Till 2 negative stool cultures obtained
- Notification: After 2005, outbreaks are notifiable
- Sanitation control:
 - Make **sanitation barrier** to ensure safe water supply and sanitation
 - Best way to control outbreak
 - Chlorination: residual chlorine level of 0.5 mg/L with a contact period of 1 hour to control cholera outbreak
- Mass chemoprophylaxis is never advisable in case of cholera outbreak as to prevent a single case of cholera, 10000 individuals must be provided with chemoprophylaxis
- Mass vaccination also not recommended

MCQ's



- Q. A 41-year-old male arrives at the emergency department with severe watery diarrhea, vomiting and abdominal cramps after travelling to a cholera outbreak region. Lab tests confirm cholera. Despite treatment he still sheds vibrio cholerae in his stool during recovery. What carrier type sheds infectious agents during recovery?
- a. Convalescent carrier
 - b. Chronic carriers
 - c. Incubatory carriers
 - d. Contact carriers

Ans: (a)



12. TYPHOID

- Gastrointestinal infection
- Typhoid is endemic in India
- Indicator of **sanitation measures** in India

EPIDEMIOLOGICAL DETERMINANTS

00:00:45

Agent	• Salmonella typhi- Gram negative, capsulated flagellar, motile bacilli • In fewer cases (3-4%)- Salmonella paratyphi
Antigens: 3 main antigens	• Somatic O antigen- Group specific • Flagellar H antigen- Type specific • Capsular Vi antigen- Virulence specific
Reservoir of infection	• Man is the only reservoir of infection
Source of infection	• Case/ carrier of typhoid
Carrier: 3 types of carriers	• Convalescent carrier: Carrier which sheds bacilli during recovery period (3 weeks- 3 months) • Temporary carriers: Sheds bacilli from 3 months to 1 year • Permanent carriers: Sheds bacilli for >1 year
Mode of transmission	• Feco-oral route ◦ Contaminated food ◦ Drinking contaminated water ◦ Fomites
Period of communicability	• Person is infectious as long as person sheds bacilli in feces or urine
Incubation period	• 10-14 days

HOST FACTORS

00:03:53

- Age: 5-19 years, M> F
- Antigen to O (Somatic) antigen is higher in **post infectious phase**
- Antigen to H (flagellar) antigen is higher in **post vaccination phase**
- Environmental factors:
 - Monsoon time(July to September)
 - Sensitive to heat, phenol, cresol

CLINICAL FEATURES

00:04:50

PHASE	FEATURES
EARLY PHASE	- Fever for 3-4 days (Stepladder pattern) - Paradoxical bradycardia - Pea soup diarrhea
LATER PHASE (7-10 DAYS)	- Splenomegaly - Rose spots (rash red bumps) on chest and trunk - Case Fatality Rate: 1-4%   Rose spots Splenomegaly
COMPLICATIONS	- Perforation of small intestine - Septicemia - Multiple organ dysfunction (MODs)- Most dangerous

DIAGNOSIS

00:06:13

- Gold standard: Bone marrow examination
- IOC (most sensitive and specific): Blood culture
- Serology: **Widal test** (least sensitive and specific)
 - Somatic O antigen: Antibodies seen in 6-8 days
 - Flagellar H antigen: Antibodies seen in 10-12 days
- Rapid serological test: Typhi dot
 - Can detect both IgM and IgG antibodies

PLAN OF INVESTIGATION (Pneumonic- BASU)

00:07:18

1 st week	Blood culture
2 nd week	Serological test (Antibody detection)
3 rd and 4 th week	3rd week: Stool culture 4th week: Urine culture

PREVENTION AND CONTROL

00:07:52

IMMUNISATION(Not in NIS)

TY21A	VI POLYSACCHARIDE VACCINE
- Live vaccine - Oral	- Subunit vaccine - SC/IM

- ≥ 5 years
- 3 dose regimen- Day 1,3,5 every 3 years
- Immunity develops after 7 days



- >2 years
- Single dose, revaccination every 3 years
- Immunity develops after 7 days



- Routinely → If there is an outbreak of typhoid → No need to vaccinate everybody as the seroconversion rate following vaccination that is antibody formation following vaccination is more than natural incubation period of the diseases
 - The best way to control typhoid in such situations is safe water supply and sanitation

VACCINE RECOMMENDATION FOR HIGH-RISK GROUPS

- Household contacts
- High-risk groups, such as hospital staff and school children
- Residents of endemic areas
- Travelers proceeding to endemic areas
- Individuals attending large gatherings
- Participants in large-scale religious ceremonies

TREATMENT

00:09:13

Isolation	• Till 3 negative stool/urine culture on 3 consecutive days
DOC	• Fluoroquinolones: 10mg/kg x 7 days
Fluoroquinolones resistance	• Can give Ampicillin/ Cefixime/ Cephalosporin
In case of carriers	• Ampicillin/ Amoxycillin+ Probenecid
Best management for carrier state	• Cholecystectomy + Ampicillin
Disinfection of stool/urine	• 5% Cresol for 2-4 hours
Soiled clothes and linen	• Chlorination and sterilization

- If there is an outbreak of typhoid:
 - **Best way to manage** → Ensure safe water supply & sanitation, achieved by chlorination
→ Chlorination- Level of residual chlorine: **0.5-0.7mg/l**
- In highly endemic areas: Raw food can be disinfected with potassium permanganate



13. SOIL TRANSMITTED HELMINTHIC INFECTIONS

SOIL TRANSMITTED HELMINTHIC INFECTIONS

INTESTINAL ROUND WORMS	<i>Ascaris lumbricoides</i>
HOOKWORMS	<i>Ancylostoma duodenale</i> , <i>Necator americanus</i>
WHIPWORMS	<i>Trichuris trichiura</i>

COMMON TYPES OF FELINE INTESTINAL WORMS



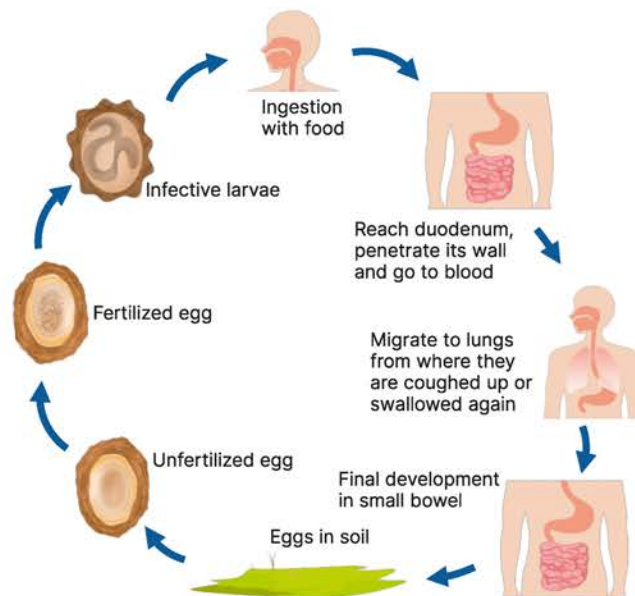
- Tapeworm → not a soil-transmitted infection

ASCARIASIS LUMBRICOIDES

00:02:00

- **M/c** soil-transmitted helminthic infection
- **Mechanism of transmission**
 - By eggs that are passed through feces of infected persons
 - Eggs released into soil where they mature and infect the host
- **Reservoir** : Humans (in small intestine)
- **Source** : Soil containing an infective form of egg
 - Females lay 2.5-3 lac eggs/day
 - Eggs become mature and infective in 2-3 weeks
- **Incubation period** : 18 days to several weeks

LIFE CYCLE



SYMPTOMS

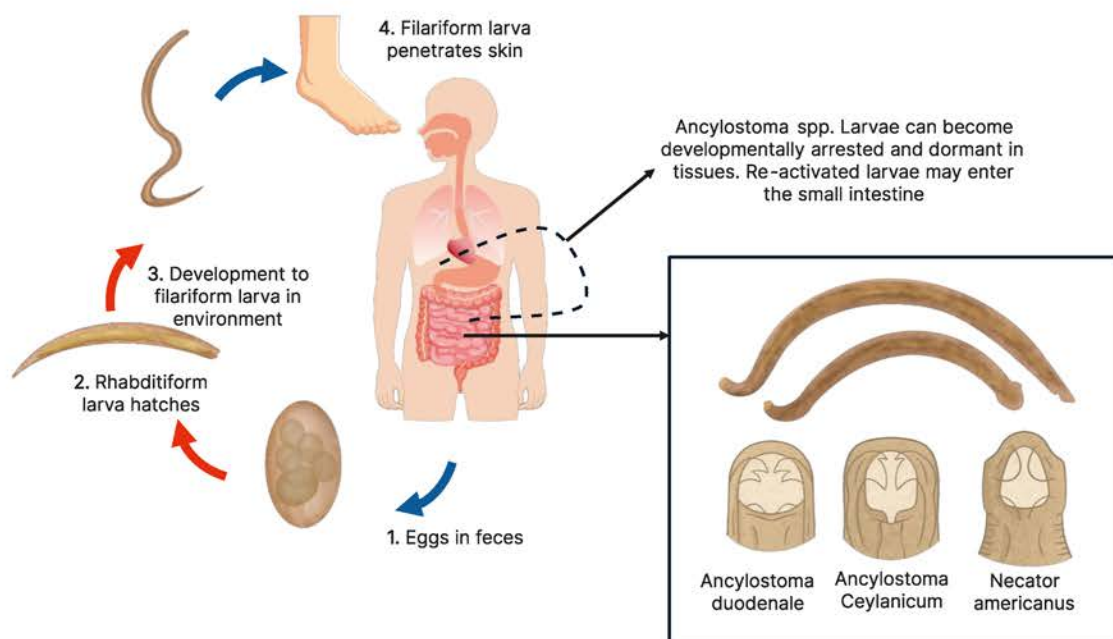
- Intestinal : Diarrhoea, abdominal pain
- Malaise, weakness
- Larva migration → Pulmonary symptoms : Asthma, cough, eosinophilia, skin rash
- GI : Intussusception, intestinal obstruction

HOOKWORMS

00:04:41

- Lives in small intestine- (*Jejunum*)
- *Man* is the only most important reservoir

LIFE CYCLE



Eggs hatch to larvae within 5-7 days (moist, porous soil) and are infective



Filariform larvae penetrate skin



Inside host, larvae migrate via lymphatics and blood to lungs and alveoli



Coughed up and swallowed again



Reach intestines where they sexually mature

EGGS PER DAY

- *Ancylostoma duodenale* : 10000 - 30000 eggs
- *Necator americanus* : 5000 - 10000 eggs

SYMPTOMS

- Chronic blood loss and iron deficiency anemia
- Hypoalbuminemia
- Responsible for child morbidity, growth retardation, LBW

PUBLIC HEALTH PROBLEM

CHANDLER'S INDEX

- Identifies the importance of hook worm Infestation as a public health problem in community
- Average number of eggs/gm of Stool
 - <200 : Not a significant public health problem
 - 200-250 : Potential danger
 - 250-300 : Minor public health problem
 - > 300 : Major public health problem

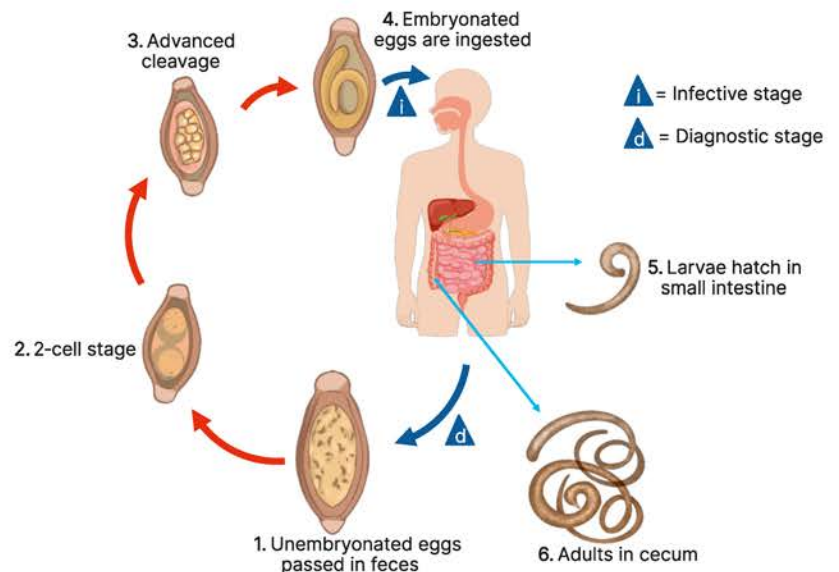
WHIPWORM

00:07:12

- Live in large intestine

LIFE CYCLE

- Females lay 200-10000 eggs per day, excreted in feces
- In soil : Eggs mature (embryonation) in 21 days
- Eggs ingested reach intestine where they hatch into larvae in the villi of small intestine
- Larva re-emerges and attaches to mucosa of colon and grows into adult



SYMPTOMS

- Mild epigastric pain, nausea, flatulence

PREVENTION AND MANAGEMENT

00:08:11

- Sanitation
- Health education
- Albendazole
- Mebendazole

NATIONAL DEWORMING DAYS

- 10th February and 10th August
- Adults : 400mg Albendazole
- Children : 200mg Albendazole



Q. How much should Chandler's index be for considering hookworm to be a minor public health problem?

- a. < 200 eggs/gm of stool
- b. 200-250 eggs/gm of stool
- c. 250-300 eggs/gm of stool
- d. > 300 eggs/gm of stool

Ans: (c)

Q. Which of the following is the national deworming day?

- a. 10th August
- b. 11th April
- c. 11th August
- d. 10th April

Ans: (a)

Q. A 12-year-old girl presents with diarrhea and abdominal pain. Her skin has a typical eosinophilic rash, she also complains of constant cough for a month. CBC shows eosinophilia. Which of the following is true about the causative agents of this infection?

- a. Source is human intestine
- b. Least common helminthic infection
- c. Reservoir is soil containing infective form of eggs
- d. Females lay 2.3 - 3 lakh eggs per day

Ans: (d)

14. DENGUE



EPIDEMIOLOGICAL DETERMINANTS

00:00:18

AGENT

- Flavivirus
- 4 serotypes: 1, 2, 3, 4
 - 1 and 2 → M/c
- No cross-immunity between serotypes is seen
 - Recovery from infection by one serotype provides lifelong immunity from that serotype only

VECTOR

00:01:54

- Aedes mosquito: *Aedes aegypti* and *Aedes albopictus*
 - Discordant feeder that requires > 1 feed: *Aedes aegypti*
 - Concordant feeder that requires no extra feed: *Aedes albopictus*
- Hunchback appearance with white stripe on black body
- AKA: Tiger mosquito
- Anthropophilic, daytime biters, and *nervous feeders* (bites multiple hosts)



- Breed in artificial collection of water
- Egg is cigar shaped, lays egg singly
- Larva: Siphon tube present - short

RESERVOIR OF INFECTION

- Man and mosquito

TRANSMISSION CYCLE

- Man – Mosquito – Man

INCUBATION PERIOD

- 3-10 days

HOST FACTORS

- All age groups - equally susceptible
- All genders - equally susceptible

ENVIRONMENTAL FACTORS

- Humidity: 60-80%
- Temperature: 16-30°C
- Does not occur at high altitude: > 2000ft
- Seen post-monsoon

CLINICAL FEATURES

00:06:17

DENGUE SYNDROME:	• Uncategorized fever: No typical features • Break bone fever: Classical dengue fever ○ Acute viral infection ○ Retro orbital pain ○ Arthralgia, myalgia ○ Bimodal presentation: peaks on day 1-2 and resurges on day 5-6 ○ Recovery within 7-10 days
Dengue Hemorrhagic Fever (DHF)	• Dengue fever + hemorrhagic manifestation (bleeding from any orifice) • Rash: Maculopapular, rubelliform type • Lab diagnosis: 20 % ↑ in hematocrit, thrombocytopenia (< 100000) • Clinical diagnosis: ○ Tourniquet test > 10 petechiae/sq inch area → probable diagnosis ○ > 20 petechiae/sq inch area → confirmed diagnosis
DENGUE SHOCK SYNDROME	• Narrowed pulse pressure (<20 mmHg) and severe plasma leakage leading to hypotensive shock and mean systolic blood pressure drops < 90 and <80 mmHg

DIAGNOSIS

00:08:06

INVESTIGATION OF CHOICE

- Within 1-5 days of fever: NS 1 (Non-Structural protein of dengue virus) antigen detection test
- > 5 days of fever: IgM ELISA antibody detection test
- Gold standard / confirmatory test: **RTPCR**

TREATMENT

00:09:11

- Fluid resuscitation
 - Crystalloid preferred
 - Initial: 0.9 % Normal saline
 - Maintenance phase: Ringer's lactate
- Paracetamol for fever, rest, blood transfusion

PREVENTION AND CONTROL

00:09:39

LARVAL SURVEILLANCE

- House or its premises searched for water-holding containers: **Sampling unit**
- Number of houses positive for Aedes larvae/ number of houses checked: **House index**
- Number of containers positive/ number of containers checked: **Container index**
- Number of containers positive for larvae / number of houses checked: **Breteau index**

LARVAL SURVEYS

- Aedes aegypti Index – House Index
 - Percentage of houses and their premises showing actual breeding of Aedes aegypti larvae
 - The index should **not be > 1%**
 - In airport and seaports in endemic areas at least **400m** around their perimeters free from breeding of Aedes aegypti

VECTOR CONTROL

00:11:30

INTEGRATED VECTOR MANAGEMENT

- Anti larval measures:
 - Environmental control (source reduction) yields permanent result
 - Chemical control: Paris green
 - Biological control
- Anti adult: indoor residual spray, fogging with malathion
- Personal protection: Bed nets
- Legislative, Implementing laws and regulations for vector control.

VACCINE

- 1st dengue vaccine: **Dengvaxia** (CYD-TDV)
- Tetravalent viral vector vaccine- freeze dried vaccine
- Diluent used is normal saline
- Route: Subcutaneous
- Dose: 0.5 ml
- Individuals 9-45 years of age in endemic areas
- Contraindications
 - Severe combined immunodeficiency (SCID)
 - HIV + person with CD4 count < 200 cells
 - Pregnant and lactating women

MCQ

00:13:57

- Q. At what age can the dengue vaccine be safely administered to individuals living in or travelling to endemic regions?
- a. 9-45 years
 - b. 8-25 years
 - c. 9-35 years
 - d. 8-50 years

Ans: **(a)**

15. MALARIA EPIDEMIOLOGY



MALARIA EPIDEMIOLOGY

00:00:24

NVBDCP

- National vector born disease control program
- Covers 6 important diseases:
 - Malaria: transmitted by Anopheles
 - Dengue: transmitted by Aedes
 - Chikungunya: transmitted by Aedes
 - Lymphatic filariasis, JE: by culex
 - Kala azar/visceral leishmaniasis: sandfly



MALARIA

- Sir Ronald Ross: demonstrated transmission of Malaria is due to a vector -Anopheles
- Burden of malaria
 - India contributes 3% of total global malaria cases
 - Africa contributes for 95% of total global malaria cases



00:03:24

EPIDEMIOLOGICAL TRIAD

AGENT

PLASMODIUM

- Plasmodium falciparum: M/c, most dangerous
- Plasmodium vivax
- Plasmodium ovale: Rare in India
- Plasmodium malariae
- Plasmodium knowlesi: rare

VECTOR

ANOPHELES

- An. culicifacies: Rural & Peri urban areas
- An. stephensi: Urban & industrial areas
- An. fluviatilis: Hilly areas, forests, forest fringes (east India)
- An. minimus: Foothills (Northeastern states)
- An. epiroticus, An. sundaicus, An. dirus: Andaman & Nicobar Islands

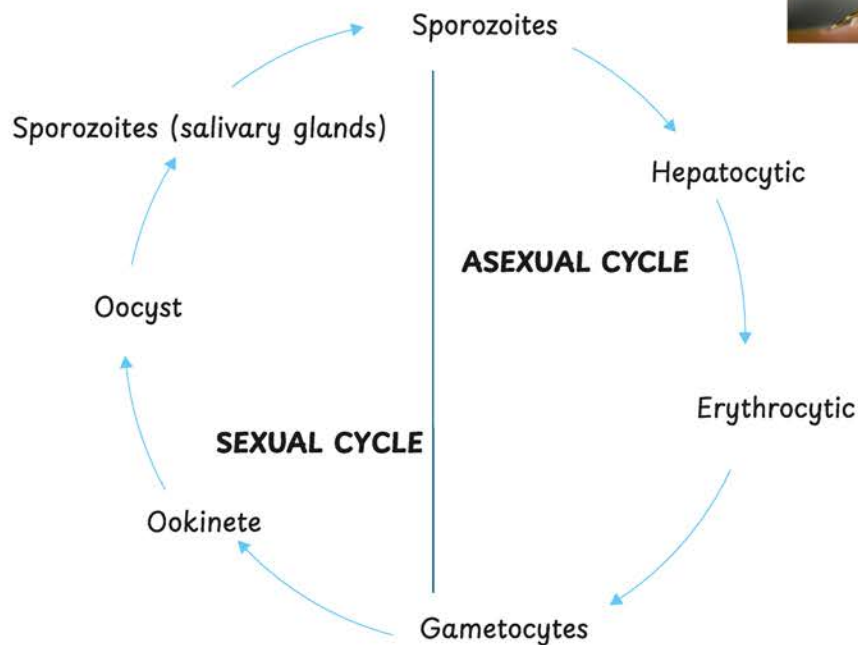
Important Information

- An. fluviatilis: Most efficient, most aggressive, works in low density
- An. culicifacies: low efficiency, require more density
- Spotted wings, sitting at an angle of 45°
- Breeds in clean water (sophisticated mosquito)

- Eggs: lay single eggs, boat shaped with lateral floats
- Larvae: no siphon tube, parallel to water surface



LIFE CYCLE



- Asexual Cycle (5-7 days): in humans – intermediate or secondary host
- Sexual Cycle (10-12 days): in mosquito – definitive or primary host
 - This 10-12 day is known as extrinsic incubation period
- In hepatocytic phase merozoites are formed
 - *P. falciparum* can produce as many as 40000 merozoites, all other species produce 10000-20000 merozoites
- In the Erythrocytic phase: Merozoite → Trophozoites → Schizont
 - Responsible for clinical features
 - In *P. malariae* this phase lasts for 72 hrs
 - Other Species: last for 36 - 48 hrs
- *P. vivax*, and *P. ovale* relapse is seen in 1-3 years
 - In hepatocytes merozoites become dormant called dormant hypnozoites, they can become active within 1-3 years
- *P. falciparum*, *P. malariae* shows recrudescence in 1-1.5 years
 - Low level erythrocytic schizogony
- Mosquitoes ingest both male & female gametocytes
 - >12 gametocytes/ mm³ of blood is present
- Man is the main reservoir

MODE OF TRANSMISSION

00:19:05

- Vector borne (m/c): Bite of infected female *Anopheles* mosquito
- Direct: Accidental injections of blood or plasma (Parasite remains infectious for at least 14 days in blood bottles stored at -4° C)

- Congenital: Rare
- Newborn infants show resistance to infection with *P. falciparum*
- Pregnancy increases the risk of malaria
- Febrile herpes is seen with malaria infections

HOST FACTORS

- Can occur in any age group
- Age < 6 months: chance of *falciparum* infection is low (fetal Hb- resistant to *falciparum*)
- Sick cell trait: resistant to *P. falciparum*
- Duffy negative RBCs: Resistant to *P. vivax*
- Holoendemic pattern: higher incidence & prevalence in 2-10 years, & at low numbers in adults

ENVIRONMENTAL FACTORS

- Season: July to November
- Malaria Awareness Month: June
- World Malaria Day: 25th April
- Higher altitudes: not seen; seen up-to 2000 - 2500 ft.
- Relative humidity: 60-80%
- Temperature: 16-34°C

INCUBATION PERIOD

- *Falciparum*: 12 days
- *Vivax*: 14 days
- *Ovale*: 16 days
- *Malariae*: 28 days

CLINICAL FEATURES

00:24:14

COLD STAGE	HOT STAGE	SWEATING STAGE
• Chills & rigors lasting for 30min -1 hour • Parasite is seen in blood • Weak pulse, fever- 100 to 104°F	• Fever- 100 to 104°F • Most uncomfortable stage, lasting for 2-6 hours • Rapid bounding pulse	• Comfortable Stage • No fever

COMPLICATION

Anemia, renal failure, hepatosplenomegaly, cerebral malaria or black water fever (*P. falciparum*), encephalitis

16. LYMPHATIC FILARIASIS



LYMPHATIC FILARIASIS

- Part of NVBDCP
- A parasitic disease

EPIDEMIOLOGICAL AGENTS

AGENT

- *Wuchereria bancrofti* : Bancroftian filariasis
- *Brugia malayi* : Malayan/brugian filariasis
- *Brugia timori* : Malayan/brugian filariasis

VECTOR

- *Wuchereria bancrofti* / lymphatic filariasis : *Culex quinquefasciatus* or *C. fatigans*
- Malayan/brugian filariasis : *Mansonia annulifera* or *M. uniformis*
- Endemic in UP, Bihar, Jharkhand, Andhra Pradesh, Orissa, Telangana, Kerala, Gujarat

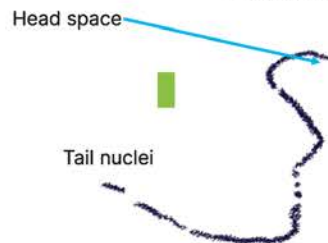
DIFFERENCES BETWEEN WUCHERERIA BANCROFTI & BRUGIA MALAYI

Wuchereria bancrofti - mf



300 x 10 μ m
Lie- Graceful curves
Head space - Short
Nuclei- Countable

Brugia malayi



260 μ
Kinked (Acute angles)
Long
Overlapping, tail nuclei+

HOST FACTORS

- Definitive host : Man (sexual cycles occurs)
- Intermediate host : Mosquito (asexual cycles occurs)
- Age: can affect any age group, peak age is 20-40 years
- Gender : M>F
- Predisposing factors : Low socioeconomic status
 - Overcrowding
 - Lack of hygiene
 - Poverty

VECTORS

- *Culex* : *Culex quinquefasciatus* or *C. fatigans* cause *Wuchereria bancrofti* / lymphatic filariasis
 - Breeds in dirty, polluted water

- *Mansonia* : *Mansonia annulifera* or *M. uniformis* cause Malayan/brugian filariasis
 - Breeds in lakes, ponds with aquatic vegetation
- Andaman & Nicobar Islands : *Aedes niveus*

MODE OF TRANSMISSION

- Bite of infected vector mosquito
- Infective stage – larval stage 3 → long thin form of larva

INCUBATION PERIOD

- Extrinsic incubation period : In intermediate host – 10-14 days
- Clinical incubation period : 8-16 months

CLINICAL FEATURES

00:09:48

- It has the following stages
 - Asymptomatic Amicrofilaremia : No symptoms, no microfilariae in blood
 - Asymptomatic microfilaria : No symptoms, microfilariae in blood
- Stage of acute manifestations
 - Occurs in first few months & years
 - Recurrent episodes of acute inflammation in lymph glands & vessels
 - Manifested as filarial fever, lymphangitis, lymphadenitis
 - Lymphoedema & epididymo - orchitis(male)
- Stage of chronic obstructive manifestations
 - Occurs 10 - 15 yrs after the onset of first acute attack
 - Causes permanent structural changes due to fibrosis & obstruction of lymphatic vessels
 - Main clinical features are hydrocele, elephantiasis & chyluria
 - Taking care of patients with elephantiasis is a tertiary level of care
 - In Brugian filariasis, genitalia are rarely involved

OCCULT FILARIASIS

00:13:36

- The classical clinical manifestations are not present
- Microfilaria is not found in the blood
- Hypersensitivity reaction to filarial antigens derived from Microfilaria
- E.g. : Tropical pulmonary eosinophilia

DIAGNOSIS

00:14:10

- Diagnostic form : Sheathed microfilariae in thick peripheral blood film
 - Direct investigation of parasite in thick peripheral blood film
 - IOC for epidemiological surveillance
- Sensitive test for detecting low density microfilaremia: membrane filtration concentration test
- Microfilariae shows Nocturnal periodicity : ↑activity at night, so demonstrate it between 10pm – 2am in blood films

SAMPLE COLLECTED

At night(10pm-2am)

- Thick peripheral blood film is prepared to demonstrate parasite
 - Due to ↑activity at night
- IOC for epidemiological surveillance
- Sensitive test: membrane filtration concentration test

At daytime

- DEC provocation test
 - Give 100 mg of DEC orally & blood examined after 1 hour
 - Investigation of parasite in thick blood films
 - Done when collection of sample is not possible at night

EPIDEMIOLOGICAL ASSESSMENT OF LYMPHATIC FILARIASIS

- M/c method used - through mass blood survey
 - Thick film using 20 cu mm of capillary blood (collected between 10pm to 2am)
- A new test for rapid diagnosis
 - FTS (filariasis test strip) : Designed to detect the antigen of *W. bancrofti* in human blood
 - Recommended & used for mapping, monitoring & evaluation activities
- Other methods
 - Most sensitive method for detecting low density microfilaremia: Membrane filter concentration method
 - DEC provocation test
 - To make *Microfilaria* appear in blood during daytime
 - 100 mg DEC given orally and blood examined 1 hour after administration
 - Clinical examination
 - Serological testing : Cannot differentiate between past & current infections
 - Xeno diagnosis : Not in India

T/t

00:21:43

- T/t for filaria (*Microfilaria* positive persons)
 - DEC : 6 mg/kg/day BD for 12 weeks
 - + Albendazole : 400 mg HS
 - + Ivermectin : 150 - 200 mcg/kg

PREVENTION AND CONTROL

00:22:29

GLOBAL LYMPHATIC FILARIASIS ELIMINATION STRATEGY

- India has set target to eliminate lymphatic filariasis by 2027, earlier it was 2020
- 3 steps done : Mapping, mass drug administration (MDA), transmission assessment survey (TAS)
 - Mapping : Measures antigenemia levels by Immunohisto- chromatography test
 - If prevalence >1% - area is endemic, start MDA
 - Mapping means conducting filarial surveys in all age groups, from 4 sentinel sites & 4 random sites, & examining 4000 slides
 - MDA : Mass drug administration - done with triple drug therapy
 - DEC(6mg/kg) + Albendazole (400mg HS) + Ivermectin(150-200mcg)
 - Given once a year for 5-6 years

- Coverage >80% of population
- Contraindicated among pregnant females and children <2 years
- TAS : Transmission assessment survey – 3 rounds (TAS-1,2,3)
- Done among 6-7 years old
- If infection threshold is <1% in each district then MDA is stopped

ELIMINATION OF LYMPHATIC FILARIASIS

- Microfilariae carrier rate : <1%
- Children born after initiation of program should be free from circulating antigenemia

DEC MEDICATED SALTS

- Started in Maharashtra
- DEC added at the rate of 1-4 gm/kg of salt

Important Information

- MDA is primary level prevention (specific protection)
- DEC Medicated Salts is also primary level prevention (specific protection)

VECTOR CONTROL METHOD

- Antilarval measures- culex
 - Environmental measures : Source reduction
 - Chemical control : Use of Paris green
 - Biologicals : Use of gambusia fish
- Mansonia : Deweeding - removal of aquatic plants, larvicidal oil

MONITORING AND EVALUATION INDICATORS

00:32:56

MICROFILARIA RATE

- Most sensitive indicator = $(\text{Persons with microfilaria} / 20 \text{ mm}^3 \text{ of blood} / \text{Total population under survey}) \times 100$
 - Indicates prevalence of infection

FILARIAL ENDEMICITY RATE

- Number of people with microfilaria + clinical diagnosis diseases / Total population under survey $\times 100$
 - Indicates prevalence of disease



Q. A group of officials from the public health department of a renowned institute conducted a filarial survey in a village with poor sanitation. For the mass blood surveys, they performed the DEC (diethylcarbamazine) provocation test on several individuals from the village. What is the correct dose of DEC administered in this test?

- a. 150 mg
- b. 100mg
- c. 200mg
- d. 50mg

Ans: (b)

Q. Which of the following states started the production of DEC- medicated salts first?

- a. Andhra Pradesh
- b. Maharashtra
- c. Madhya Pradesh
- d. Tamil Nadu

Ans: (b)

Q. A 44-year-old male presents to the community health center complaining of swelling in his lower extremities. A blood test confirms the presence of microfilariae. The patient is diagnosed with lymphatic filariasis, considering the t/t options, which drug is not included in the triple drug therapy regimen for filariasis?

- a. DEC(Diethylcarbamazine)
- b. Ivermectin
- c. Doxycycline
- d. Albendazole

Ans: (c)



17. KALA AZAR

KALA AZAR

00:00:15

- **Black sickness:** Pigmented hands, face, feet and abdomen
- AKA: Oriental boils, Baghdad boils, kala-azar, black fever, sandfly disease, dum-dum fever, white leprosy, espundia

EPIDEMIOLOGICAL DETERMINANTS

00:00:38

AGENT

Visceral leishmaniasis (kala-azar)	• <i>Leishmania donovani</i>
Cutaneous leishmaniasis (oriental sore)	• <i>Leishmania tropica</i>
Mucocutaneous leishmaniasis	• <i>Leishmania braziliensis</i>

- Endemic in Bihar, UP, West Bengal, Jharkhand, Orissa
- *L. donovani* is a **digenetic parasite**: Completes its life cycle in 2 host
 - Vertebrate host(humans): **amastigote form** → LD bodies
 - Insect(sandfly): **promastigote form** → flagellated form

Important Information

- India had set 2020 to be target year to eliminate kala azar
 - Elimination criteria: <1 case /1000 population
 - Now has set target to eliminate kala azar by 2027

RESERVOIR OF INFECTION

- Dogs, Jackals, foxes, rodents and mammals
- **Indian kala-azar: Non zoonotic infection**

HOST FACTOR

- All age groups are equally susceptible
- Peak age: 5-9 years
- M/c in males
- Rural areas, agricultural areas, overcrowding, poverty, malnutrition

ENVIRONMENTAL FACTORS

- Rain, winter, early spring

VECTOR

- Female Sandfly: Anthropophilic

- **Phlebotomus argentipes**: Kala-azar (visceral leishmaniasis)
- *P. papatasi*: Cutaneous leishmaniasis
- *P. sergenti*: Mucocutaneous leishmaniasis
- Features
 - Hops -50 yards
 - Lanceolate-shaped, hairy wings, long legs
 - Smaller than mosquitoes
 - Breed in cool, damp, moist places: crack and crevice of tree holes
- Geographical area of distribution: rural > urban



KEY POINT

- Insecticide for sandfly: **DDT** (sprayed up to height 6 ft from floor) 1-2 gm/sq m
- Insecticide of choice: **pyrethroids**

INCUBATION PERIOD

- In humans: 1-4 months
- Extrinsic incubation period (sandfly): 1-4 weeks

Important Information

- Active case finding is done in kala azar → Multipurpose worker search for cases every fortnightly

CLINICAL FEATURES

00:10:25

- **Kala-azar/visceral leishmaniasis**
 - Fever - **irregular, double rise in temperature**
 - Weight loss
 - Anemia,
 - Hepatosplenomegaly (not responding to anti- malarial t/t)
- **Cutaneous leishmaniasis**
 - Ulceration and nodules on skin
- **Mucocutaneous leishmaniasis**
 - Severe form of Cutaneous leishmaniasis
 - Involve oral mucosa, leading to facial disfiguration
- **Post kala-azar dermal leishmaniasis: PKDL**
 - Immune response of body to kala-azar
 - Presence of nodules filled with LD bodies in skin

DIAGNOSIS

00:14:09

Rapid diagnostic kit (rk39)	Method of choice for diagnosis of kala Azar & epidemiological survey
Confirmatory test	Demonstration of parasite LD bodies in aspirates of liver, spleen, bone marrow, lymph nodes or skin
Aldehyde test of Napier	Good for surveillance

Montenegro test	- Becomes positive after 2-3 months of disease onset and reverts to negative 6 months after cure of Leishmaniasis - Distinguishes immune from non-immune subjects
Haematological test	- Leukopenia, - Anemia, - Reversed albumin-globulin ratio ↑ IgG and WBC: RBC ratio

T/T

00:15:20

- Single dose IV infusion of **Liposomal Amphotericin B (LAMB)** for all age groups
 - Dose: 10mg/kg body weight, single dose
- Other drugs
 - Miltefosine:** 10mg - 50mg in the age group 2-65 years for 28 days
 - +
 - Paromomycin:** 11-12 mg/kg weight for 10-11 days
- Miltefosine can be given alone
- PKDL: Miltefosine 100mg/day for 12 weeks

VECTOR CONTROL

- Insecticide of choice Pyrethroids

OUTCOMES

00:17:44

- No signs and symptoms of kala-azar + complete t/t: **cure**
- No signs and symptoms after 6 months of completed t/t: **final cure**
- Persistence of signs and symptoms even after 2 weeks of t/t: **failure/no response**
- Signs and symptoms within 6 months of t/t completion: **relapse**

INDICATORS

00:18:27

- Detection rate = $\frac{\text{Total cases detected}}{\text{Total population}} \times 100$
- Complication rate = $\frac{\text{Number of patients completed t/t}}{\text{Number of patients started on t/t}} \times 100$
- Cure rate = $\frac{\text{Number of patients cured}}{\text{Number of patients started on t/t}} \times 100$
- Failure rate = $\frac{\text{Number of failure or non response}}{\text{Number of patients started on t/t}} \times 100$
- Operational indicators of program:
 - Detection rate
 - Cure rate



- Q. Which of the following is not a feature of the vector that is responsible for transmitting the infectious agent causing kala azar?
- a. Sandfly is the vector
 - b. Sandfly is identified by hairy wings
 - c. *Phlebotomus papatasi* and *P. sergenti* cause visceral leishmaniasis
 - d. They live in cracks and crevices of trees/cool, damp places

Ans: (c)

- Q. Which of the following areas is not endemic to kala azar?
- a. West Bengal
 - b. UP
 - c. Bihar
 - d. Assam

Ans: (d)

18. JAPANESE ENCEPHALITIS



EPIDEMIOLOGY

00:00:18

AGENT

- Group B arbovirus
 - Family: Flavivirus
 - Single-stranded RNA

VECTOR

- **Culex**: *C. tritaeniorhynchus*, *C. vishnui*
- Breeds in dirty, polluted water, irrigated rice fields, shallow ditches, long-standing water collections
- Mode of transmission: bite of vector

ENDEMIC

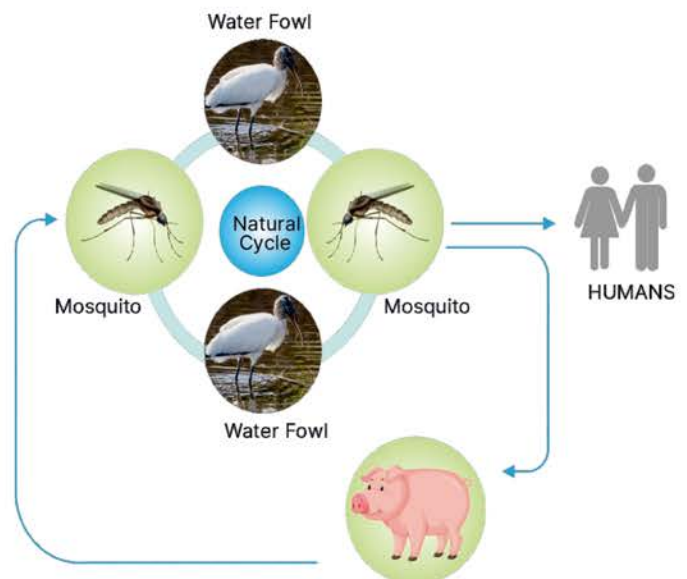
- UP, Bihar, Jharkhand, Assam, West Bengal, Haryana

HOST FACTORS

- **Pigs are amplifier host**: Maintain viremia, but no sign and symptoms
- **Birds** (Ardeid birds, pond herons, ducks, poultry): Natural reservoir – **maintenance host**
- Cattle & buffaloes: Mosquito attractants
- Man: Dead-end host/ accidental host
- Horses: Only animal that presents with symptoms
- Age: 85% infections occur < 15 years
 - Not common in infancy

LIFE CYCLE

- Life cycle of JE virus is:
 - Pig - mosquito - pig
 - Ardeid bird - mosquito - ardeid birds





Culex mosquito: hunchback, no white stripe on body, short body, long legs

KEY POINTS

- Not all humans bitten by mosquitoes develop disease
- Ratio of JE overt disease to inapparent infection varies from 1:300 to 1:1000
 - Follows **iceberg phenomenon**

ENVIRONMENT

- Seasonal distribution
 - M/c - Post Monsoon: rainy season, paddy farming
- Culex requires temperature of 16°C - 34°C
- Humidity: 60-80%

INCUBATION PERIOD

- 6-8 days

CASE FATALITY RATE

- 15-30%, Avg 20%

CLINICAL FEATURES

00:07:45

PRODROMAL STAGE	• Lasts for 1-7 days • Presents with fever, coryza, malaise, fatigue
ACUTE ENCEPHALITIS STAGE	• Lasts for 3-7 days • Presents with convulsions, neurological deficits, altered sensorium, hypotonia
POST ENCEPHALITIS STAGE	• Residual neurological deficits seen • In late-stage death can occur, seen in 9-12 days

DIAGNOSIS

00:09:50

- IgM ELISA: within 5-7 days
- Confirmatory is RT-PCR

TREATMENT

00:10:10

- Symptomatic

PREVENTION

00:10:17

VACCINATION

- Live & Killed vaccine can be used under NIS → interchangeable

LIVE VACCINE	KILLED VACCINE
• Strain: SA-14-14-2 → Cell culture vaccine • Schedule: ○ 1st dose – 9 months (JE₁) ○ 2nd dose: 16-24 months (JE₂) • Dose: 0.5 ml, Subcutaneous route at upper left arm • Does not follow open vial policy	• Since 2023, government of India is promoting killed vaccine as well • Killed Vero cell culture vaccines ○ JEEV: strain- SA-14-14-2 ○ JENVAC: strain - KOLAR • Dose: 0.5 ml, intramuscular route • Site: Anterolateral aspect of left thigh • Follow open vial policy • Better immunogenicity than live vaccine • Schedule: JE₁– 9 months, JE₂ – 16-24 months

VECTOR CONTROL

00:14:15

- Done according to Integrated vector management
 - Antilarval measure:
 - Source reduction
 - Chemical control – Paris green
 - Biological methods
 - Anti-Adult methods: fogging
 - Personal measures (most imp) : mosquito nets

MCQ

Q. Amplifier host of Japanese encephalitis?

- a. Pig
- b. Dogs
- c. Cats
- d. Birds

Ans: (a)



19. YELLOW FEVER AND KFD

YELLOW FEVER

00:00:22

- Zoonotic disease, prevalent in Africa and South America
- Exotic to India (Does not occur in India)
- Occurrence of a single case of yellow fever in India will be called as Epidemic

EPIDEMIOLOGY

00:01:52

AGENT

- Yellow fever virus : Arbovirus belongs to family of flavivirus

RESERVOIR

- Man, primates (monkeys), mosquitoes are vectors, not reservoirs

VECTOR

- *Aedes aegypti*

PERIOD OF COMMUNICABILITY

- Man is infective for 3 - 4 days of illness
- Mosquitoes are also infective after completing the IP of 8 - 10 days
- Once infective the mosquito shall remain so all its life



HOST FACTORS

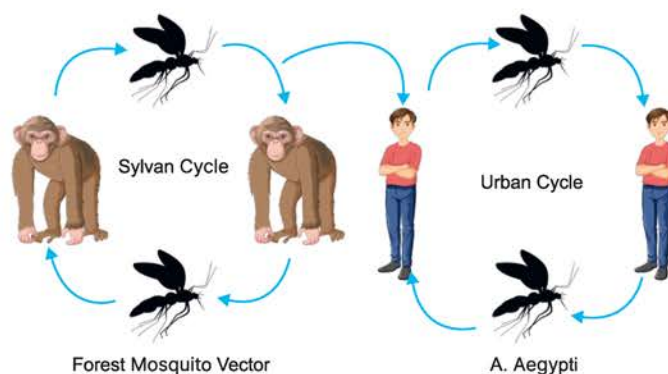
- All age groups and genders are equally susceptible
- High-risk group: People working in forests

ENVIRONMENTAL FACTORS

- Temperature 24 - 36° C
- Relative Humidity > 60%
- One attack gives lifelong immunity

LIFE CYCLE

00:04:27



- **Sylvatic** (jungle cycle) : Monkey - mosquito - man
- **Urban** cycle : Man - mosquito - man
- Incubation period in Man : 3-6 days
- Extrinsic Incubation period (mosquitoes) : 8-10 days
 - **Transovarian spread is seen in mosquitoes**

CLINICAL FEATURES

00:06:13

- Hemorrhagic fever (black vomit, epistaxis, Melena), hepatic and renal manifestations
- CFR : Up to 80% in severe cases

CAN YELLOW FEVER OCCUR IN INDIA?

- Yes
- India is yellow fever receptive area
- Favorable conditions for transmission exist: Population unvaccinated and susceptible
- Vector *Aedes aegypti* found
- Climate conditions are favorable
- Reservoir: Monkeys are susceptible

DIAGNOSIS

00:07:10

- Serological : IgM specific
- Confirmatory : RT-PCR (Reverse Transcription Polymerase Chain Reaction)

TREATMENT

00:07:24

- Symptomatic

VACCINE

00:07:40

- Strain : **17 D vaccine**
- Storage : -15 to -30° C (< 0° C)
- Type of vaccine : live attenuated vaccine, freeze-dried / reconstituted vaccine
- Diluent used : Cold saline
- Route : Subcutaneous
- Dose : 0.5ml
- Immunity starts after 5-7 days of vaccination
 - **Validity of vaccination certificate begins 10 days after vaccination and remains lifelong**
- Contraindications : Egg allergy, immunocompromised individuals, children < 6 months
- Can be given in pregnancy

VECTOR CONTROL

00:10:23

- **Aedes aegypti index** : Number of houses showing breeding of Aedes larvae within 400 m of airports and seaports
 - **This AAI is kept as <1% (or zero) within 400 m of airports and seaports**
- Quarantine : A traveler without valid international certificate of vaccination is placed on quarantine in a mosquito proof ward for 6 days

LARVAL SURVEY

00:12:07

- House/premise index (HI)

$$HI = \frac{\text{no. of house (+) for Aedes species}}{\text{no. of house inspected}} \times 100\%$$

- Container index (CI)

$$HI = \frac{\text{no. of containers (+) for Aedes sp}}{\text{no. of containers house inspected}} \times 100\%$$

- Breteau index (BI)

$$HI = \frac{\text{no. of positive containers}}{\text{total no. of houses inspected}} \times 100\%$$

KYASANUR FOREST DISEASE (KFD) VIRUS ECOLOGY

00:12:40

- Zoonotic disease : monkey disease
- 1st seen in Shimoga, Karnataka

EPIDEMIOLOGY

AGENT

- KFD virus : Arbovirus of family flavivirus
 - Previously Group B Togavirus
 - Togaviruses and Flaviviruses were once classified together but are now separate families

INCUBATION PERIOD

- 3 - 8 days

CFR

- CFR: 5-10%

VECTOR

- Hard tick (Haemaphysalis spinigera and Haemaphysalis turturis in India)
- Transmitted to man by bite of infective ticks (especially nymphal stages)
- The chitinous covering is called **scutum**



Large Scutum
(Male)



Small Scutum
(Female)

MAIN RESERVOIR

- Small mammals : Rats, squirrels

AMPLIFYING HOSTS

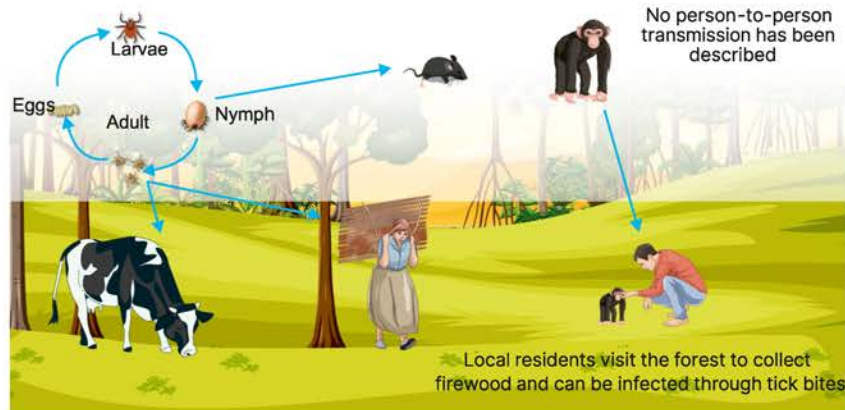
- Monkeys and cattle
- Humans are accidental hosts

HOST FACTORS

- Majority of cases : 20 - 40 years
- Males > female
- Mostly occurs in people working in forest

LIFE CYCLE

00:14:56



- Infective stage : Nymphal stage (Cattle)
- This nymphal stage infects rats, monkeys
- No human-to-human transmission

CLINICAL FEATURES

00:15:27

- Early stage : Fever, GI symptoms
- Hemorrhagic stage : Fever, bleeding manifestation
- Afebrile stage : 2-3 weeks
- **Meningoencephalitis stage** : neurological deficits

CONTROL

00:16:31

- Control ticks : 5% Malathion, 3-5% BHC
- Insecticide spray (carbaryl, fenthion, propoxur - 2.24kg / hectare in hot spot (50 m around the spot of monkey death) besides endemic foci
- Restriction of cattle movement in forest
- Vaccination of at-risk population
- Personal protection : Adequate clothing, insect repellent (dimethyl phthalate)



Q. Which of the following vaccines should not be given with yellow fever vaccine?

- a. Cholera vaccine
- b. Tetanus vaccine
- c. Hepatitis B vaccine
- d. Varicella vaccine

Ans: (a)

Q. Which among the following is the case fatality rate of Kyasanur forest disease?

- a. 5-10%
- b. 20-30%
- c. 40-60%
- d. 80-90%

Ans: (a)

20. RABIES



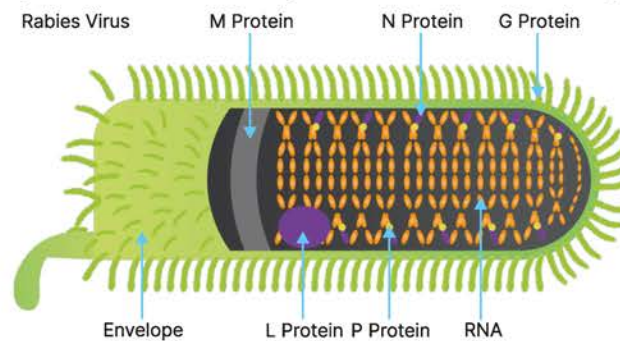
RABIES

- Zoonotic disease
- Case fatality rate: 100%
- Does not follow iceberg phenomenon
- Management in India is covered under National rabies control program

EPIDEMIOLOGY

AGENT

- Lyssavirus Type 1 (Bullet shaped RNA virus) belongs to Rhabdoviridae family



TWO TYPES OF VIRUSES

FEATURES	STREET VIRUS	FIXED VIRUS
Source	Isolated from infected samples: Natural case of rabies	Isolated after Serial passage in rabbit's brain
Pathogenicity	Always pathogenic	Not pathogenic
I.P	Variable (1-3 months)	Fixed (5 - 10 days)
Negri Bodies	Present	Absent
Comments	Produce disease	Produce vaccine
Salivary glands	Present	Absent

Important Information

- Negri bodies : Spherical cytoplasmic inclusions in neurons, consist of rabies virus proteins & RNA
- Hallmark of rabies encephalitis

HOST

- Age : 1 - 25 years
- M > F

ENVIRONMENTAL FACTOR

- Based on exposure
 - Outdoor : forest officers, wildlife photographers
 - Indoor : veterinarians, dog handlers

ROUTES OF TRANSMISSION

00:07:22

- Animal bites (dogs, cats, monkeys, cow, goat, sheep, buffalo, horses)
 - Mouse bite or rat bite not associated with rabies
- Licks : on abraded skin or abraded/unabraded mucosa
- Aerosols : Rabies infected bats
- Person to person : human to human transmission of rabies through bites or saliva is theoretically possible but till now no documented case
- Transplantation : documented cases of human-to-human transmission have been seen in people receiving organ transplants from infected donors
- No evidence that rabies can be transmitted by drinking milk from an infected animal

INCUBATION PERIOD

- 2-8 weeks

CFR

- 100%

KEY POINT

- M/c form of rabies in India : urban rabies (associated with dog bite)
- Form of rabies not reported in India : bat rabies
- Rabies is not found in
 - Australia, China (Taiwan), Cyprus
 - Iceland, Ireland
 - Malta, Japan
 - New Zealand, Britain
 - Andaman and Nicobar Islands (India)
 - Lakshadweep (India)

MANAGEMENT OF RABIES

00:12:43

- Wound management
- Rabies vaccine
- Rabies immunoglobulins

CATEGORIES OF CONTACT

CATEGORIES OF CONTACT WITH SUSPECTED RABID ANIMAL	POST-EXPOSURE PROPHYLAXIS (AFTER ANIMAL BITE)
Category I • Touching of feeding animals • Licks on intact skin	None

Category II • Nibbling of uncovered skin • Minor scratches or abrasions without bleeding	• Local t/t of wound • Anti rabies vaccine
Category III • Single or multiple transdermal bites or scratches • Licks on broken skin • (Bleeding present) • Contamination of mucous membrane with saliva from licks • Contacts with bats	• Local t/t of wound • Immediate vaccination • Administration of rabies immunoglobulin

WOUND MANAGEMENT

PHYSICAL	Wash with running tap water	Mechanical removal of virus from wound
CHEMICAL	Wash the wound with soap & water, Apply disinfectant	Inactivation of virus
BIOLOGICAL	Infiltrate immunoglobulin in the depth & around the wound in category III exposures	Neutralization of virus

MANAGEMENT OF ANIMAL BITE

- Local wound t/t
 - Purpose : remove as much as virus possible from peripheral nerve endings before it reaches brain
→ done by flushing or washing wound area with running tap water & soap for >15 to 20 minutes
→ Virucidal agents can be used : alcohol, tincture iodine, povidone iodine
 - Suturing : immediate suturing not recommended, if required it is done 24-48 hours later
 - Reduce chances of rabies by 80%

RABIES VACCINATION

POST EXPOSURE PROPHYLAXIS REGIMENS

- Updated THAI red cross regimen (recommended by National rabies control program of India)
 - Intradermal (I/D) regimen
 - Dose : 0.1 ml
 - 4 visits : on day 0 (1st dose of vaccine), day 3, day 7, day 28
 - Given as 2-2-2-2 : on 2 sites simultaneously, so total 8 dose each 0.1ml
 - Adult site : deltoid, in children : anterolateral aspect of thigh
- ESSEN regimen (given in private set up) : Intramuscular (I/M) regimen
 - Dose : 0.5 ml
 - 5 visits : On day 0, day 3, day 7, day 14, day 28
 - Given as 1-1-1-1-1, on 1 site so total dose is 5 each 0.5 ml

RE-EXPOSURE GUIDELINES

- Dog bite within 3 months
 - Nothing needs to be done

- > 3 months
 - 2 visits : day 0, day 3
 - Either I/M or I/D regimen

PRE- EXPOSURE GUIDELINES

- Reserved for high-risk groups : forest officers, wildlife photographers, vets, dog handlers
- 3 visits : day 0, day 7, day 21/28
- Either I/M or I/D regimen

OTHER SCHEDULES

- One week : 2 sites : ID regimen : Institute Pasteur du Cambodge regimen : 2-2-2-0-0
- Three weeks : IM regimen : 5 doses, ZAGREB regimen : 2-0-1-0-1

RABIES IMMUNOGLOBULIN (RIG)

00:31:01

- Anti Rabies serum
 - Human Rabies Immunoglobulin : preferred in 20 IU/ kg, I/M route
 - Maximum 70% of Ig has to be administered around the wound, remaining at a site distant from the wound
 - Equine Ig : 40 IU/ kg : given only when human immunoglobulin is not available
 - I/M route, 50% around the Wound, 50% at site away from wound
- Criteria to give RIG
 - Only administered for post exposure prophylaxis
 - Only once in a lifetime
 - Given in all class 3 bites
 - For class 2 animal bites (immunocompromised individual – HIV+)
 - Can be given up to 7 days from 1st dose

SPECIAL CONSIDERATIONS

- Any contraindication to post exposure prophylaxis : No, as rabies is 100% fatal
- Rabies vaccine in pregnancy, HIV, Immunocompromised: Killed vaccine
- Individuals on chloroquine t/t for malaria : IM regimen preferred; ID regimen not preferred due to reduced rate absorption
- Rabies Immunoglobulin can be given in Cat II exposure : Immunocompromised (HIV +ve)
- Criteria for rabies booster
 - If VNA (Vaccine neutralizing antibodies) level < 0.5 IU : booster dose given every 2-3 years
- Suffered a dog bite from a vaccinated pet dog : take vaccination according to category of wound
- Dog remains healthy on being observed for 10 days : never stop vaccination

ANTI RABIES VACCINE

00:38:30

- Nervous Tissue Vaccines- obsolete now
- Vero cell culture vaccine
- Primary Cell Culture Vaccines : Human Diploid Cell Vaccines
- Vaccines approved for ID use : Verorab, Rabipur, Abhayrab

Important Information

- Rabies vaccines are freeze dried vaccines (powdered form); Diluents used
- Do not follow open vial policy, need to be used within 4-6 hrs

RABIES IN DOGS

00:39:44

- To control rabies in dogs
 - Elimination of stray & ownerless dogs along with prophylactic vaccination of dogs against rabies
 - Most logical & cost-effective approach to control rabies as 80% of dog population in an area is accessible
 - Primary dose of vaccine given at 3-4 months, & boosters at regular intervals

RABIES IN DOGS MANIFEST IN 2 FORMS

- Furious rabies : mad dog syndrome
 - Change in behavior
 - Running amuck
 - Change in voice
 - Excessive salivation
 - Paralytic stage
- Dumb rabies : paralytic form, die in 3 days
- Incubation in dogs: 3-8 weeks

CONTROL OF RABIES IN DOGS

00:43:39

- Prophylactic vaccination

TYPES OF VACCINES

- BPL inactivated nervous tissue vaccine (single dose)
 - Dose : 5ml for dogs & 3ml for cats
 - Revaccination advised every 6 months & subsequently every year
- Modified live virus vaccine:
 - Dose is 3 ml (single dose) & boosters every 3 years

OTHER METHODS INCLUDE

- Registering & licensing of all domestic dogs
- Restraint of dogs in public places
- Immediate destruction of dogs & cats bitten by rabid animals
- Quarantine for 6 months of imported dogs
- Health education of people regarding care of dogs & prevention of rabies

LABORATORY DIAGNOSIS

- Cut the head of the animal & send it to the nearest lab covered in ice or send a brain in 50% glycerol solution
- Fluorescent antibody test (FAT) : detect viral antigen in brain, most rapid and best
- Microscopic examination : of brain tissues for Negri bodies
- Intracerebral mouse inoculation
- Corneal test: rabies vaccine can be demonstrated in corneal impression of live animals or skin biopsy of live animals by FAT
 - It does not rule out infection



- Q. An urban area has a higher number of rabies cases, the most cost- effective and logical approach to control rabies is?
- Testing all dogs for rabies
 - Capacity building
 - Removing stray dogs and vaccinating dog population
 - Building cooperation among health workers

Ans: (c)

- Q. Which among the following is the region where vaccination according to Essen regimen is given in adults?
- Gluteal region
 - Deltoid region
 - Subscapular region
 - The anterolateral aspect of the brain

Ans: (b)

- Q. Which of the following is the place in India where Rabies is not found?
- Lakshadweep
 - Karnataka
 - Sikkim
 - Kashmir

Ans: (a)

- Q. Which of the following is the dose of human rabies immunoglobulin?
- 20 IU/kg IM
 - 40 IU/kg IM
 - 30 IU/kg IM
 - 50 IU/kg IM

Ans: (a)

- Q. Which of the following are true about different categories of rabies bites?
- Category 1 : bleeding occurs
 - Category 2 : licks or touch on intact skin
 - Category 3 : contamination of mucus membrane from the saliva through licks
 - Category 4 : require immediate rabies vaccination

Ans: (c)

21. LEPTOSPIROSIS



EPIDEMIOLOGY

00:00:11

AGENT

- *Leptospira interrogans* : Spirochetes, gram -ve bacteria, long helical coiled, thin & motile

RESERVOIR

- Rodents (*Rattus norvegicus*, *Mus musculus*), rabbits : Definitive host
- Intermediate host : carrier species (cattle, horses, sheep, goats)

MODE OF TRANSMISSION

- Direct contact
- Indirect contact : Contact of broken skin with soil, water, vegetation contaminated with *Leptospira*
- Droplet infection : Inhalation while milking cows
- IP : 4-20 days

Important Information

- For leptospirosis 3 buzz words : rats, rice fields, & rain

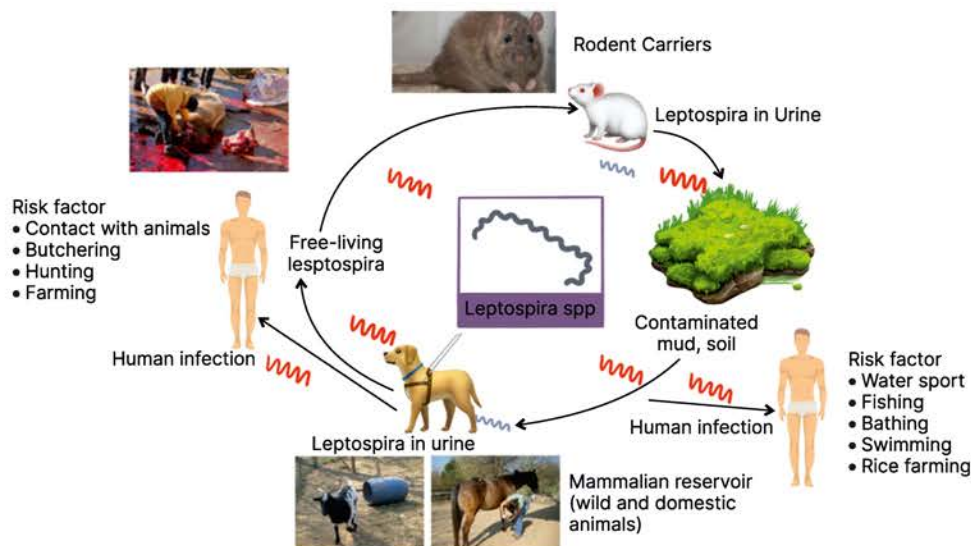
HOST FACTORS

- Males > Females
- M/c age: 20-45 years
- Urbanization, water sports, swimming, river rafting

ENVIRONMENTAL FACTORS

- M/c zoonotic infection reported after floods
- In floods there is ↑sed water logging
- Urbanization (construction of dams, canals) : rodents come out of burrows
- ↑ soil alkalinity, temperature of soil < 10° C

LIFE CYCLE



CLINICAL FEATURES

00:05:52

- Anicteric leptospirosis
 - Milder form
 - Pulmonary manifestations are primary concern
- Icteric leptospirosis
 - More severe form
 - Jaundice, fever, myalgia, hypotension, circulatory failure
- Pregnancy : Fetal complications, abortions

DIAGNOSIS

00:06:10

- Microscopic examination of blood, urine, CSF
 - Blood smear : Dark field examination
 - IgM ELISA : Sensitive test for diagnosis
- Screening test : MAT – (Microscopic agglutination test)
- Diagnostic test / IOC : NAT- (Nucleic acid test)

T/T

00:07:09

- DOC : Penicillin 6 million units daily I/V
- Alternate drugs : Tetracycline, Amoxicillin, Ampicillin

PREVENTION AND CONTROL

00:07:17

- Rodent control : Spray zinc phosphide into burrow
- Health education
- Prophylaxis : Doxycycline 200mg/week for 6 weeks
 - Impending outbreaks, highly endemic area

MCQ

- Q. All are true about leptospirosis, except?
- a. Leptospira is released into the rodent's urine
 - b. The rodent urine contaminates the mud/ soil or water
 - c. Males are affected mostly
 - d. The m/c age group is 15-25 years

Ans: (d)

22. PLAGUE

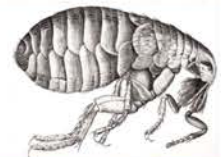


- Zoonotic disease with epidemic potential
- **Notifiable disease**
- **Other names** - Black death (gangrene formation)/Mahamari/The Great death

EPIDEMIOLOGY OF PLAGUE

00:01:29

Agent	• Yersinia pestis - Gram negative, non-motile, coccobacillus • Identification ◦ Bipolar staining with Wayson's stain ◦ Giemsa stain → also used • Reservoir of infection: Wild rodent - mice, field rats, Indian Gerbil (Tatera indica) • Source of infection ◦ Infected rodents ◦ Rat flea ◦ Case of pneumonic plague
Host factors	• Predisposing factors ◦ Overcrowding ◦ Lack of hygiene ◦ Travellers
Environmental factors	• Transmission season: September - May • Temperature: 16° - 34° Celsius • Relative humidity > 60%
Incubation Period	• Pneumonic Plague: 1 - 3 days • Bubonic Plague: 2 - 7 days • Septicaemic Plague: 2 - 7 days
Mode of Transmission	• M/c / Principle / Most efficient mode of transmission: Vector - Rat flea (Xenopsylla cheopis) → Blood sucking arthropod ◦ Both male & female rat flea bite and transmit disease ◦ Rat flea ingests 0.5 mm³ of blood/blood meal → 5000 plague bacilli present • Blocked flea (choked flea) → Propagative transmission i.e., Plague bacilli multiply in number → transmit disease • Partially blocked flea - more dangerous in transmitting disease (↑ lifespan → Survive longer)



Xenopsylla cheopis (oriental rat flea) engorged with blood

Flea Index

- **Total flea index:** Total number of fleas/rat
- **Cheopis index:** Total number of cheopis/rat
 - Specific flea index
 - Most sensitive criteria during plague
 - Increases during plague
 - **Cheopis index >1** → explosiveness of plague
- **Burrow index:** Total number of fleas/burrow
- **Earliest sign** of upcoming plague: ↓ in number of rats

CLINICAL FEATURES

00:13:30

→ Bubonic Plague

- M/c (80-85%)
- Incubation period: 2 to 6/7 days
- Rat Flea bites on the lower limbs → inoculation of bacilli → intercepted by local lymphatics → formation of Bubos
- Fever, chills, headache, myalgia, vomiting, diarrhoea, painful lymphadenitis
- No person-to-person transmission
- Case fatality rate: 50-60% (if untreated)
- Complications: Pneumonic Plague & Septicemic Plague

→ Pneumonic Plague

- Rare
- Most hazardous
- Person to person transmission seen (highly communicable)
- Case fatality rate: 100%. (if not aggressively treated)
- Primary or secondary manifestation
- Incubation period: 1-3 days
- Complication of Bubonic plague or droplet infection from pneumonic plague

→ Septicemic Plague

- Rarest
- Complication of Bubonic plague: Widespread dissemination from lymph node of Bubonic plague
- Accidental lab infection
- Very severe
- No person - to - person transmission
- Incubation period: 2-7 days
- Eschar formation
- Gangrene of cooler parts: Finger, toes, ear & nose – black death
- Collapse & multiorgan failure
- Case fatality rate: 100% (if untreated)
- Dead end; rarely spread to close contacts possible

DIAGNOSIS

- Staining
 - **Wayson's stain** → Bipolar staining
 - Giemsa stain
- Blood culture - **Gold standard for diagnosis**
 - Cary Blair transport media → isolate bacilli
- Serology: (Ab) demonstration

CONTROL OF PLAGUE

- Diagnosis
- Notification (must)
- Isolation of cases → 7-10 days
- Quarantine period (for healthy contacts) → 40 days
- Treatment
 - **Streptomycin** → 30 mg/kg BD x 7 to 10 days
 - **Tetracycline** → 30-40 mg/kg OD x 7 to 10 days

PREVENTION & MANAGEMENT

- **Most effective method** to break transmission of plague: **Rat fleas reduction via using insecticides**
- **Vector Control**
 - 10% DDT
 - 3% BHC
 - Alternate: 2% Carbaryl, **5% malathion (M/c used)**
- Flea index <1 or 0 within 48 hours of spraying → Vector control

PREVENTION

- **Vaccination:** Effective if carried out 1 week before anticipated plague outbreak
- Vaccine strain: **Sokhey modified Haffkine vaccine**
- 2 doses
 - Day 0: 0.5ml, SC
 - Day 14: 1ml SC, (day 7 to 14)
- Immunity developed 5 to 7 days after vaccination
- Booster dose for high-risk individuals at 6 monthly intervals
- Not effective during outbreaks
- Effective in emporiatrics (Travellers)

CHEMOPROPHYLAXIS

- Surveillance of contacts for 7 days
- Tetracycline / Doxycycline 500 mg OD x 5 days



Q. Which among the following is the most recommended insecticide to control rat fleas?

- a. 5% Malathion
- b. DDT
- c. BHC
- d. Carbaryl

Ans. (a)

Q. What is the dose of the Sokhey – modified Haffkine vaccine?

- a. Day 0 – 0.5ml SC, Day 14 – 1ml SC
- b. Day 0 – 0.1ml SC, Day 5 – 0.5ml SC
- c. Day 0 – 0.3 ml SC, Day 14 – 1ml SC
- d. Day 0 – 0.5 ml SC, Day 5 – 0.5ml SC

Ans. (a)

Q. A 20-year-old male came to the hospital complaining of swelling in the groin for 1 week. On examination, Bubos were seen on the groin and lower limb. Which among the following is the antibiotic used for chemoprophylaxis?

- a. Tetracycline
- b. Chloramphenicol
- c. Penicillin
- d. Erythromycin

Ans. (a)

23. TETANUS



EPIDEMIOLOGY

00:00:14

AGENT	• <i>Clostridium tetani</i> ◦ Gram positive anaerobic drumstick-shaped spore-forming bacilli
CAUSE	• Exotoxin
TYPES	• 2 types – Tetanolysin & Tetanospasmin • Tetanospasmin ◦ Most dangerous and most potent toxin ◦ Responsible for opisthotonus (lockjaw) ◦ Blocks the release of presynaptic inhibitory neurotransmitter
SOURCE AND RESERVOIR	• Same, i.e., soil
MODE OF TRANSMISSION	• Contamination of wound with spore-forming bacilli • Droplet • Transplacental • Contact inoculation
PERIOD OF COMMUNICABILITY	• Not contagious → no transmission from person to person ◦ No period of communicability ◦ Spores can survive in the environment for number of years
INCUBATION PERIOD	• 6-10 days & can extend upto months

NEONATAL TETANUS

00:04:37

- Neonatal Tetanus / Tetanus Neonatorum / **8th-day disease**
- M/c cause: Infection of umbilical stump
- Home deliveries → Application of unclean stuff or blade on umbilical cord → Neonatal tetanus
- Neonatal tetanus elimination: India eliminated neonatal tetanus in 2015 (September 2015)
- Criteria for elimination: **<1case/1000 live births**
- Most important reason for elimination – TT vaccination of pregnant women



	NT INCIDENCE PER 1000 LIVE BIRTHS		COMPLETE TD COVERAGE (2 OR MORE DOSES)		CLEAN DELIVERY
NT HIGH-RISK AREAS	>1	or	<70%	or	<50%
NT CONTROL AREAS	<1	and	>70%	and	>50%
NT ELIMINATION AREAS	<1	and	>90%	and	>75%

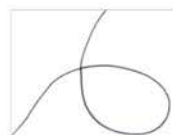
PREVENTION OF MATERNAL AND NEONATAL TETANUS

00:07:17

- Safe, clean delivery
- **7 cleans**
 - Clean surface
 - Clean hands
 - Clean cord
 - Clean cut
 - Clean tie
 - Clean water
 - Clean towel
- Timely vaccination of pregnant women

Clean hands of
attendants

Clean surface

Sterile cutting
instrument

Clean tie



Clean cloth



Clean towel

PREVENTION AND CONTROL

ACTIVE IMMUNIZATION

TD VACCINE IN NATIONAL IMMUNIZATION SCHEDULE

- 6 weeks: Penta 1
- 10 weeks: Penta 2
- 14 weeks: Penta 3
- 16 - 24 months - 1st DPT booster
- 5 - 6 years - DPT 2nd booster
- 10 years: Td 1
- 16 years: Td 2
- A total **7 doses** of tetanus are administered

IN PREGNANCY

Primi or multi with last childbirth > 3 years ago

- 2 doses of Td vaccine:
 - 1st dose: as soon as possible
 - 2nd dose: 4 weeks later

Multi with last childbirth within 3 years

- Booster dose of Td: 7th month of pregnancy

PASSIVE IMMUNIZATION

00:10:15

- Human tetanus immunoglobulin
- **Dose:** 250 IU for all age groups, and effectivity lasts for 30 days

For a woman who has not received a single dose of the Td vaccine during pregnancy

↓ Administer

Antitoxin 750 IU within 6 hours of birth to baby



Important Information

- CDC recommends administration of single dose of Tdap vaccine to pregnant women between 27 - 36 weeks of gestation.
- Also protects against pertussis and diphtheria

TETANUS MANAGEMENT PROTOCOL FOR TRAUMA CASES

00:13:08

CATEGORY	CLEAN WOUND	UNCLEAN WOUND
A: Full immunization within 5 years	Only wound care	Only wound care
B: Full immunization > 5 years age - < 10 years	Wound Care + TT single dose	Wound Care + TT single dose
C: Full immunization > 10 years ago	Wound Care + TT single dose	Wound Care + TT single dose + Human Tetanus immunoglobulin
D: Unknown, never taken immunization	Wound Care + TT complete course	Wound care + TT complete course + Human Tetanus Immunoglobulin



Important Information

- Active and passive Immunization is given in the case of
 - Tetanus
 - Rabies
 - Hepatitis B
 - Diphtheria
- Active and Passive immunization together is not given in Measles

ANTIBIOTICS IN TETANUS

00:15:00

- Non-immune person with sustained injury
- Not active against spores
- Only active against vegetative form of bacilli
- 1.2 mega units of Benzathine penicillin → within 6hrs of injury

MCQ

Q. Which Tetanus vaccine is given for trauma cases?

- TT Vaccine
- Td Vaccine
- DPT Vaccine
- Pentavalent Vaccine

Ans (a)

24. HIV / AIDS & STD's

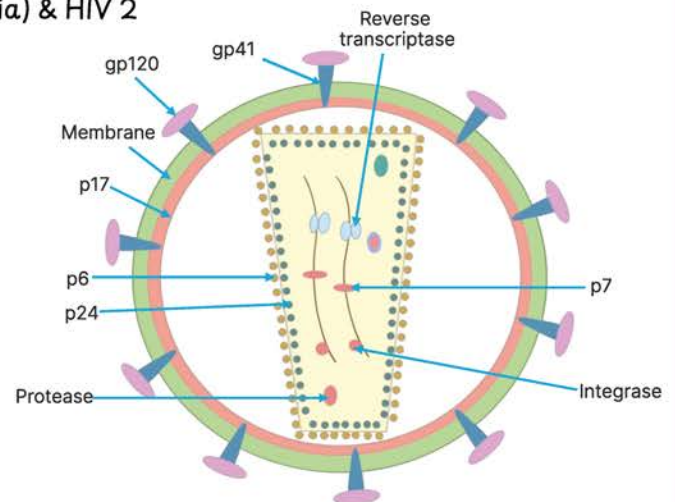


00:00:32

EPIDEMIOLOGICAL DETERMINANTS

AGENT: Single-stranded RNA virus

- **Family:** Lentivirus
- **Two types:** HIV 1 (M/c in India) & HIV 2



- HIV is
 - Sensitive to alcohol, solvents, & detergents
 - Resistant to radiations

ROUTE OF TRANSMISSION

- Most efficient route of transmission: **blood & blood products**
 - Which route has maximum efficiency, i.e., maximum risk of transmission
- M/c route of transmission of HIV/AIDS: sexual route **heterosexual > homosexual**
 - Which route contributes the maximum burden

ROUTE OF TRANSMISSION	PREVALENCE	EFFICIENCY OF ROUTE
Sexual transmission	>87%	0.01 - 1%
Through Blood & Blood products	<1 %	90%-95%
Sharing needles & syringes	<1%	5-10%
Needle stick injury	< 1%	0.3%
Vertical (Mother to child transmission)	5%	In developed countries 25% In developing countries 30%

HOST FACTORS	• Age: M/c age group 30- 44 years ◦ HIV-afflicted age groups in India: → 30- 44 years > 15- 29 years > 45+ years > under 15 years • Gender: Affects both genders								
HIGH RISK OF TRANSMISSION	• The risk of transmission is more from males to females (because the area of exposure is more) ◦ The chance of transmission from semen is more than vaginal secretion • Anal intercourse chance of transmission is more than vaginal intercourse • In the presence of other STD infection → risk of transmission ↑ 10 times • Adolescents are at a higher risk of transmission due to thin mucosal lining • Urine: viral load is low • Saliva: viral load is negligible								
SEGMENTS OF POPULATION COVERED	<table><tr><th>HIGH RISK GROUPS/ POPULATION</th><th>BRIDGE POPULATION</th><th>GENERAL POPULATION</th></tr><tr><td> • CSW • MSM • Injectable drug users • Transgenders </td><td> • Clients of CSW • STD infection • Single male migrants • Truck drivers </td><td> • ANC clinics → converted to Sentinel sites → biennial surveillance ◦ This type of surveillance is known as Sentinel surveillance • Prevalence among antenatal women is used as a proxy indicator for the general population </td></tr></table>	HIGH RISK GROUPS/ POPULATION	BRIDGE POPULATION	GENERAL POPULATION	• CSW • MSM • Injectable drug users • Transgenders	• Clients of CSW • STD infection • Single male migrants • Truck drivers	• ANC clinics → converted to Sentinel sites → biennial surveillance ◦ This type of surveillance is known as Sentinel surveillance • Prevalence among antenatal women is used as a proxy indicator for the general population		
HIGH RISK GROUPS/ POPULATION	BRIDGE POPULATION	GENERAL POPULATION							
• CSW • MSM • Injectable drug users • Transgenders	• Clients of CSW • STD infection • Single male migrants • Truck drivers	• ANC clinics → converted to Sentinel sites → biennial surveillance ◦ This type of surveillance is known as Sentinel surveillance • Prevalence among antenatal women is used as a proxy indicator for the general population							
TARGETED INTERVENTIONS (TI)	• Specific groups of interventions directed toward high-risk groups • E.g., ◦ Providing female condoms to CSW under NACP ◦ Opioid substitution therapy								
EPIDEMIOLOGICAL PATTERN OF HIV EPIDEMIC IN INDIA:	Epidemic starts from → Commercial sex workers, homosexuals, drug users - highest risk group ↓ Clients of sex workers, STD patients, migrant population, partners of drug users - bridge population • This type of epidemic is known as a concentrated epidemic • The general population (pregnant females taken as proxy indicators after taking informed consent)								

CLASSIFICATION OF STATES/ UTS	GROUP I (HIGH PREVALENCE):	• Maharashtra, Tamil Nadu, Andhra Pradesh, Karnataka, Manipur, Nagaland • High-risk groups: > 5% • Antenatal clinics: > 1%		
	GROUP II (MODERATE PREVALENCE):	• Gujarat, Goa, Pondicherry • High-risk groups: > 5% • Antenatal clinics: < 1%		
	GROUP III (LOW PREVALENCE):	• Remaining states & UTs • High-risk groups: < 5% • Antenatal clinics: < 1%		
CATEGORIZATION OF DISTRICTS	DISTRICT CRITERIA	ANC	HIGH RISK GROUPS (HRG)	HOTSPOTS
	CATEGORY A DISTRICTS	>1% among antenatal women anytime anywhere in the last 3 years	-	-
	CATEGORY B DISTRICTS	<1%	>5% among HRG anytime anywhere in the last 3 years	-
	CATEGORY C DISTRICTS	<1%	<5%	Known
	CATEGORY D DISTRICTS	<1%	<5%	With no known hotspots
STAGES OF HIV	Stage 1 Asymptomatic	Stage 2 Mild disease	Stage 3 Moderate disease	Stage 4 Severe disease (AIDS)
	• No symptoms • Or only persistent Generalized Lymphadenopathy	• Wt. loss>5-10% • Sore or cracks around the lip	• Wt. loss>10% • Oral thrush • Oral hairy Leucoplakia	• HIV wasting syndrome • Esophageal thrush

		• Seborrhea • Prurigo • Herpes zoster • Recurrent URTI • Recurrent mouth ulcer	• More than 1 month ◦ Diarrhea ◦ Unexplained fever ◦ Severe bacterial infection ◦ Pneumonia ◦ Muscle infection • Pulmonary TB • TB lymphadenopathy • Acute necrotizing ulcerative gingivitis	• More than 1 month: ◦ Herpes simplex ulceration • Lymphoma • Kaposi sarcoma • Invasive cervical cancer • Pneumocystis pneumonia • Extrapulmonary TB • Cryptococcal meningitis • Toxoplasma brain abscess • Visceral leishmaniasis • HIV encephalopathy
	HIV: Human Immunodeficiency Virus, AIDS: Acquired immunodeficiency Syndrome			
	Important Information • People with HIV are given ART irrespective of CD4 count • M/c opportunistic infection among PLHIV in India is TB > Candida • M/c cause of death in PLHIV in India is TB • M/c opportunistic infection among PLHIV in the world is Pneumocystis jirovecii (formerly called Pneumocystis carinii) Pneumonia			

STD

SYNDROMIC MANAGEMENT OF STI

00:22:16

- Method of treating STI (Sexually Transmitted Infection) based on symptoms rather than laboratory confirmation
 - To control STIs (Endorsed by WHO)

STRATEGIES

- To identify syndromes: Group symptoms and signs into syndromes
- Providing treatment: Treat the most common organisms causing syndromes
- Use flowcharts: Guide decision making
- Counselling patients: Educating patients about the disease and encourage them to take treatment and use condoms
- Follow up: Follow up with patients to ensure their response to treatment

KIT	COMPOSITION	TARGETTED SYNDROME
Kit-1 (Grey)	Tab. Azithromycin (1000mg) & Tab. Cefixime (800mg) single dose	• Urethral discharge syndrome • Vaginal discharge syndrome (For Cervicitis) • Painful scrotal swelling • Presumptive treatment
Kit-2 (Green)	Tab. Secnidazole (2000mg) & Tab. Fluconazole (150mg) single dose	• Vaginal discharge syndrome (For Vaginitis)
Kit-3 (White)	Inj. Benzathine penicillin G (2.4MU) & Tab. Azithromycin (1000mg) single dose & disposable syringe 10ml with 21gauge needle and sterile water 10ml	• Genital ulcer disease syndrome (For Syphilis and chancroid)
Kit-4 (Blue)	Tab. Doxycycline 100mg (28 capsules BD for 14 days) & Tab. Azithromycin 1g single dose	• Genital ulcer disease syndrome (For Syphilis and Chancroid when unavailability or history of allergy to BPG)
Kit-5 (Red)	Tab. Acyclovir 400mg (21 tablets TID for 7 days)	• Genital ulcer disease syndrome (For Herpetic ulcers)
Kit-6 (Yellow)	Tab. Cefixime 800mg single dose & Tab. Metronidazole 400mg (28 tablets BD for 14 days) & Cap. Doxycycline 100mg (28 capsules BD for 14 days)	• Lower abdominal pain • Pelvic Inflammatory Disease
Kit-7 (Black)	Tab. Doxycycline 100mg (42 capsules BD for 21 days)	• Inguinal bubo under genital ulcer disease syndrome • LGV proctitis under anorectal discharge syndrome
Kit-8 (Brown)	Tab. Cefixime 800mg stat dose & Tab. Doxycycline 100mg (14 capsules BD for 7 days)	• Anorectal discharge syndrome

- STI clinics - Suraksha clinics under National AIDS Control Programme

BENEFITS

- Prompt treatment (Treatment at first visit)
- High cure rates (Immediate treatment)
- Low cost (Laboratory investigations not required)
- Accessible (Non STI specialist can treat)



- Q. Which of the following is considered a Bridge population in Sentinel surveillance for HIV under the National AIDS Control program?
- a. Injectable drug abusers
 - b. Transgenders
 - c. Single male migrants
 - d. Health workers

Ans: (c)

- Q. A 47-year-old male patient presents to the community health centre with a recent diagnosis of HIV. He is concerned about the possibility of transmitting the virus to his partner & asks for information regarding the modes of transmission & risk factors associated with HIV. Based on this scenario, which of the following statements about the transmission of HIV is NOT true?
- a. The concentration of HIV is higher in semen than in vaginal secretions
 - b. The viral load is higher in Saliva
 - c. The viral load is lesser in urine
 - d. The risk of transmission is higher from male to female

Ans: (b)



25. RICKETTSIAL DISEASES

RICKETTSIAL DISEASES AGENT

- Rickettsiae : Gram -ve, non-spore forming, pleomorphic, obligate intracellular organism

DISEASES	RICKETTSIAL AGENT	VECTOR	MAMMALIAN HOST
TYPHUS GROUP			
Epidemic typhus	R. prowazekii	Louse	Humans
Murine typhus/ Endemic typhus	R. typhi	Flea	Rodents
SPOTTED FEVER GROUP			
Indian tick typhus	R. conorii	Tick	Rodents, dogs
Rocky mountain spotted fever	R. rickettsii	Tick	Rodents, dogs
Rickettsial pox	R. akari	Mite	Mice
ORIENTIA GROUP			
Scrub typhus	O. tsutsugamushi	Trombiculid mite (larvae: 3 pairs of legs)	Rodents
OTHERS			
Q fever	C. burnetii	Nil	Cattle, Sheep, goats
Trench fever	R. quintana	Louse	Humans

- Scrub typhus clinical features
 - Fever, rash on 5th to 7th day of fever, lymphadenopathy
 - Eschar formation : Black spot near bite
 - Investigation : Weil Felix reaction
- Q fever has no vector in humans, it is transmitted by inhalation of dust or ingestion of contaminated meat

T/t

- Doxycycline



Q. Which among the following is the vector of Q fever?

- a. No vector
- b. Mite
- c. Louse
- d. Tick

Ans: (a)

Q. Which among the following is the rickettsial agent of Rickettsial pox?

- a. *R. rickettsii*
- b. *R. akari*
- c. *R. conorii*
- d. *R. typhi*

Ans: (b)



26. EBOLA, NIPAH AND ZIKA VIRUS

00:00:14

EBOLA

- Disease of the African subcontinent
- Belongs to Bioterrorism Category A

Family

Flaviviridae

Serotypes

- Zaire (M/c) - Ebola
- Reston
- Tai
- Bundibugyo

Reservoir of infection

Bats and wild animals

Source of infection

Case of Ebola

Incubation Period

2-21 days

Modes of transmission

Bat bite, sexual route, body secretions (semen, vaginal secretion, blood, feces, urine)

Case Fatality rate

50-90%

Clinical Feature

Fever, rash, nausea, vomiting, pneumonia, bleeding diathesis

Complications

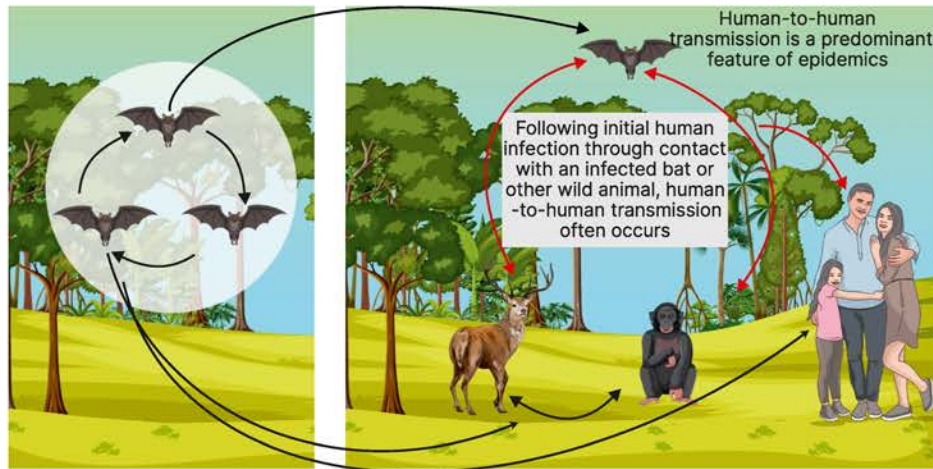
Renal and liver failure

Diagnosis

- ↑ Liver enzymes,
- thrombocytopenia
- RTPCR (Confirmatory) > ELISA

Prevention & Control

- T/t: remdesivir, monoclonal antibodies
- Interrupt modes of transmission by sanitation
- Control - Vaccine: viral vector vaccine
 - Efficiency: 50-60%
 - Given to high-risk groups

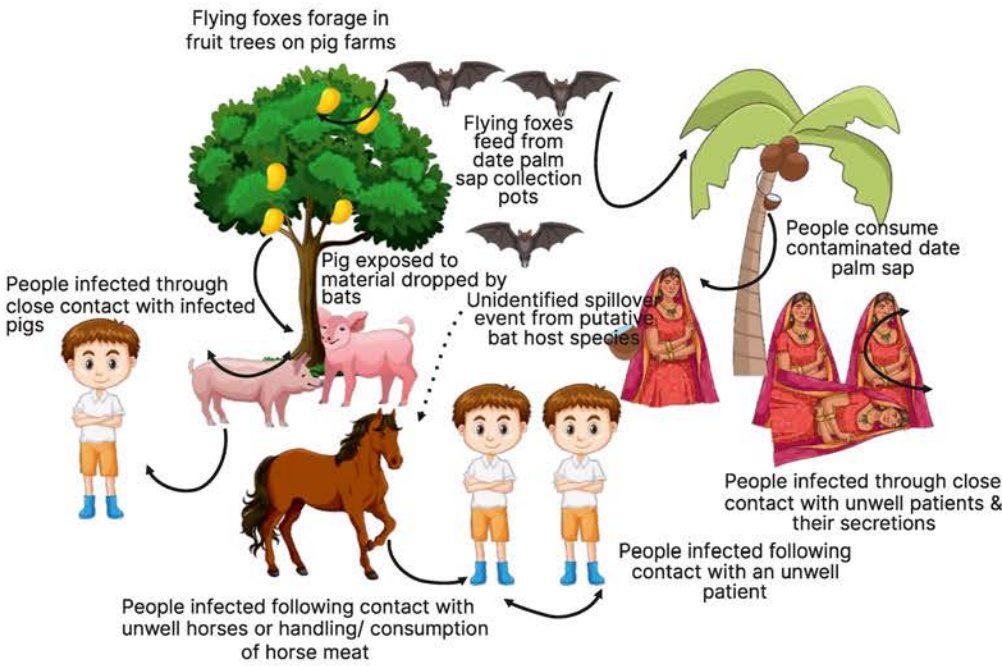


NIPAH

00:05:53

- Belongs to the Bioterrorism Category C
- Emerging/ reemerging infection
- Epidemics in Kerala: 2018, 2021, 2023

FAMILY	• Paramyxoviridae
SEROTYPES	• B serotype: Spotted in Bangladesh • M serotype: Spotted in Malaysia
ROUTE OF INFECTION	• Large fruit bats (genus: Pteropus)
INTERMEDIATE HOST	• Pigs (India), horses, cattle
SOURCE OF INFECTION	• Infected humans & animals
MODES OF TRANSMISSION	• Bats → humans via ingestion of infected fruits • Human to human via respiratory route i.e., droplet infection • Animals → humans via contact
INCUBATION PERIOD	• 4-14 days
QUARANTINE	• For healthy contacts - 21 days
CASE FATALITY RATE	• 40% - 70%
C/F	• 30-40% - Asymptomatic • High grade Fever, diarrhoea, vomiting
COMPLICATIONS	• ARDS (pneumonia , myocarditis, encephalitis (M/c), behavioural change

DIAGNOSIS	• RTPCR > ELISA
TREATMENT & CONTROL	• Screen all family members ◦ Wear PPE to take throat swab, blood, CSF, urine • Symptomatic treatment • Ribavirin • Monoclonal antibodies
DISCHARGE CRITERIA	• RTPCR - negative on 2 different occasions, 5 days apart • After discharge, the person should be under observation for 21 days
LIFE CYCLE	 Flying foxes forage in fruit trees on pig farms Flying foxes feed from date palm sap collection pots People infected through close contact with infected pigs Pig exposed to material dropped by bats Unidentified spillover event from putative bat host species People consume contaminated date palm sap People infected through close contact with unwell patients & their secretions People infected following contact with unwell horses or handling/ consumption of horse meat People infected following contact with an unwell patient

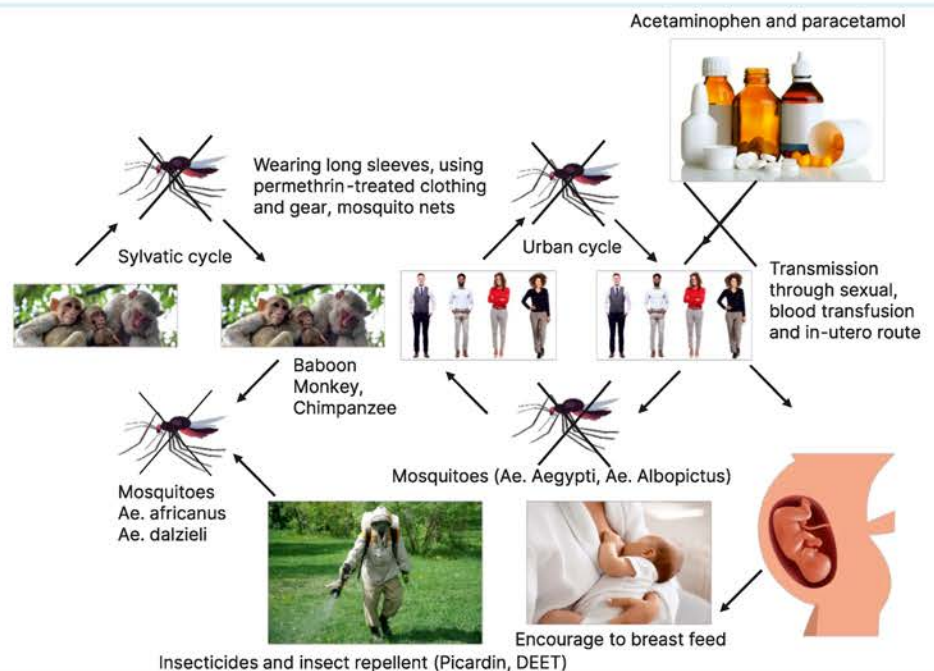
ZIKA

00:12:52

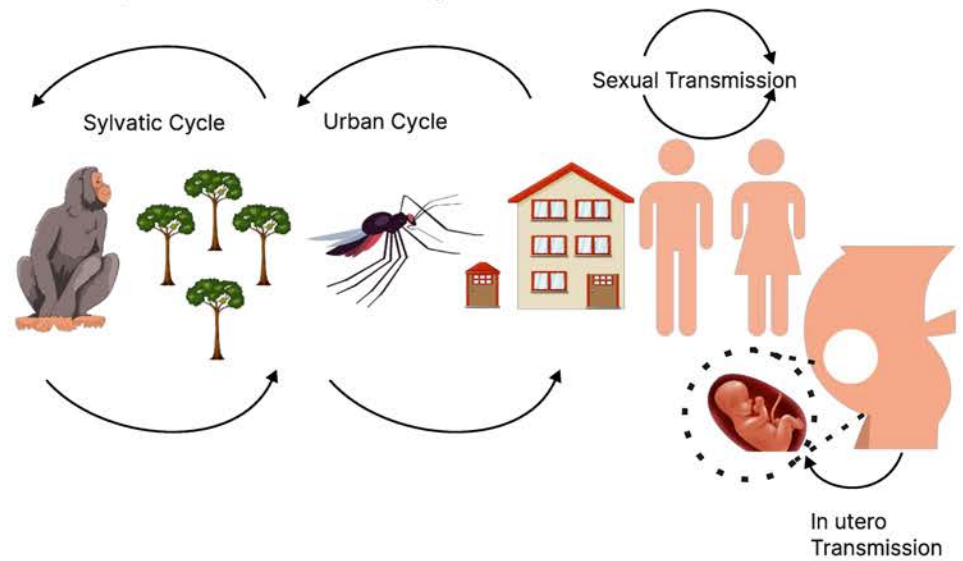
- In 2015 → declared as a public health emergency of international concern
- In 2018 → more cases were seen in Rajasthan

AGENT	• SS RNA virus
FAMILY	• Flaviviridae
RESERVOIR OF INFECTION	• Monkey, chimpanzee, wild animals
SOURCE OF INFECTION	• <i>Aedes aegypti</i> • <i>Aedes albopictus</i>
MODES OF TRANSMISSION	• Monkeys → Mosquitoes → Humans • Vertical transmission: Mother → Child ◦ Zika congenital syndrome: microcephaly, limb atrophy, visual/ hearing loss, stillbirth
INCUBATION PERIOD	• 5-15 days
C/F	• 50-60% - asymptomatic • Self-limiting illness • Fever • Rash • Conjunctivitis • Arthralgia • Myalgia
COMPLICATION	• GBS (Guillain Barre Syndrome) • Transverse myelitis
DIAGNOSIS	• RTPCR
CONTROL	• <i>Aedes aegypti</i> index <1%

LIFE CYCLE



- Jungle cycle: Chimpanzee, *Aedes* mosquito
- Urban cycle: Human, *Aedes* mosquito





Q. Zika virus is transmitted by:

- a. *Aedes aegypti*
- b. *Culex*
- c. *Anopheles*
- d. *Phlebotomus papatasii*

Ans: (a)

Q. Mark the incorrect statement regarding the Nipah virus.

- a. It is transmitted through the bat's urine
- b. It causes haemorrhagic fever
- c. It is present in India
- d. It is an emerging infection

Ans: (c)